

Photonic Cooling Microrefrigeration

A thermodynamic argument for laser-driven on-chip refrigeration and the Photonic Cold Plate.

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TECHNICAL PRESENTATION

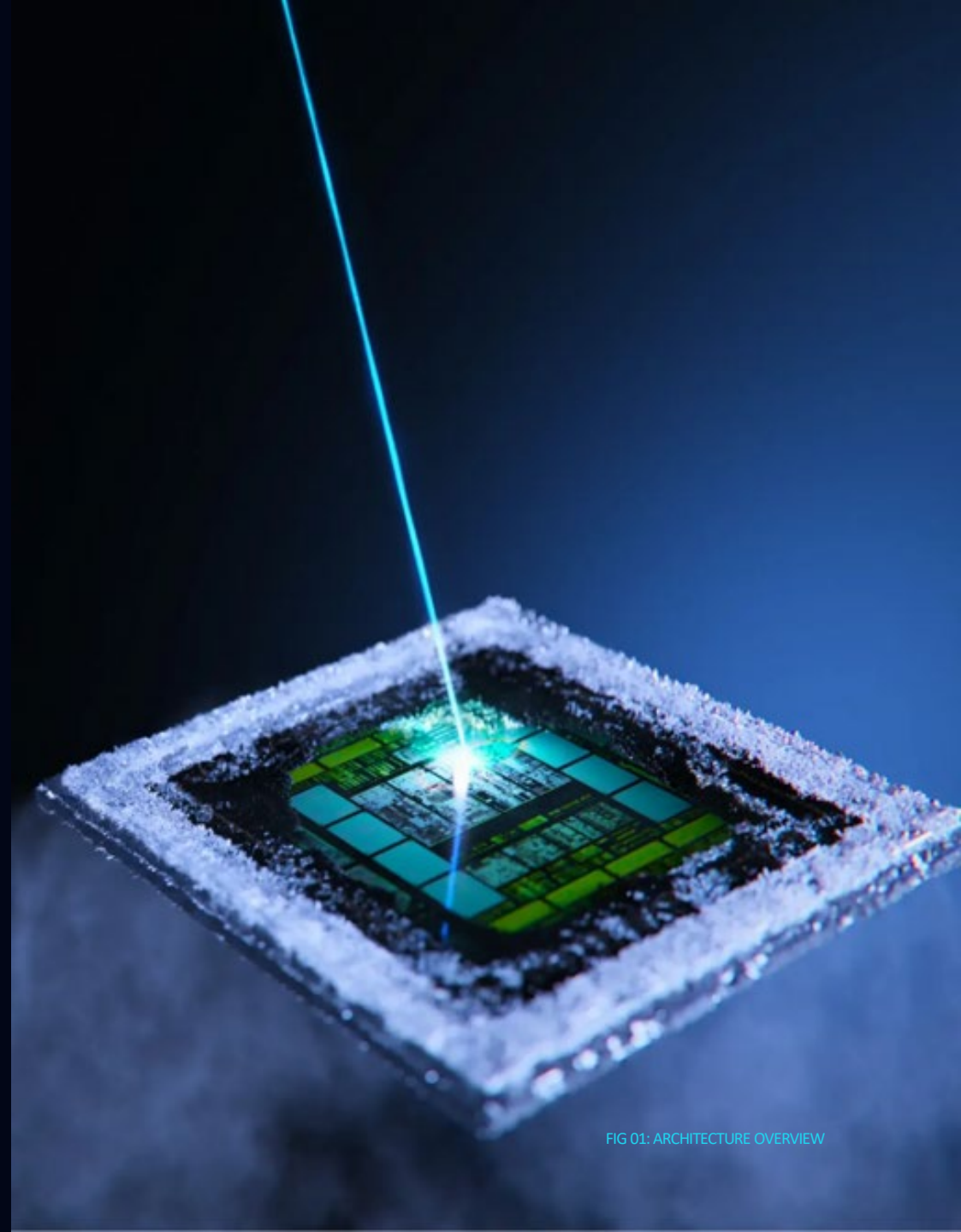
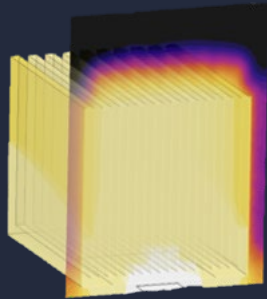


FIG 01: ARCHITECTURE OVERVIEW

The Power Wall: Chip Heat Flux Has Reached Conduction Barrier

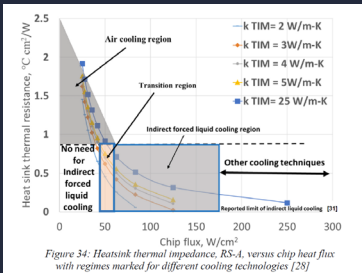
The Power Wall

Air cooling fails above ~1 W/mm². The thermal resistance of the air-interface becomes too high.



Microfluidic ceiling

Manifold microchannels reach ~10 W/mm². For context, the Sun's surface is ~63 W/mm². Modern chips already approach 10–15% of that intensity.



Local COP matters

Macro systems chase global COP. Micro systems care about local COP — moving high q with little pumping work.

Heat-flux trajectory across silicon eras

Era	Typical heat flux (W/mm ²)	Primary cooling method
1950s–60s · Discrete transistors	< 0.01	Natural convection / air
1980s · Early CMOS / VLSI	0.01 – 0.1	Passive aluminum heat sinks
2000s · Pentium 4 / multi-core	0.1 – 0.5	Active air, heat pipes
2020s · H100 / H200-class AI	0.5 – 2.0	Vapor chambers / direct liquid
2025+ · SiC/GaN, 3D-stacked	5 – 10+	Micro-channels / jet impingement

Refs: ITRS Heterogeneous-Integration Roadmap 2023 · Nature Comm. 15, 4275 (2024) – manifold μ -channels

TODO: The Conduction bottleneck in practice (hot spots vs nominal power), Dennard Scaling, Thermal limits to chip design, transistor density, latency, memory, etc... Dark Silicon

Cooling Has Bifurcated: Macro Efficiency vs. Micro Power Density

Two divergent trajectories

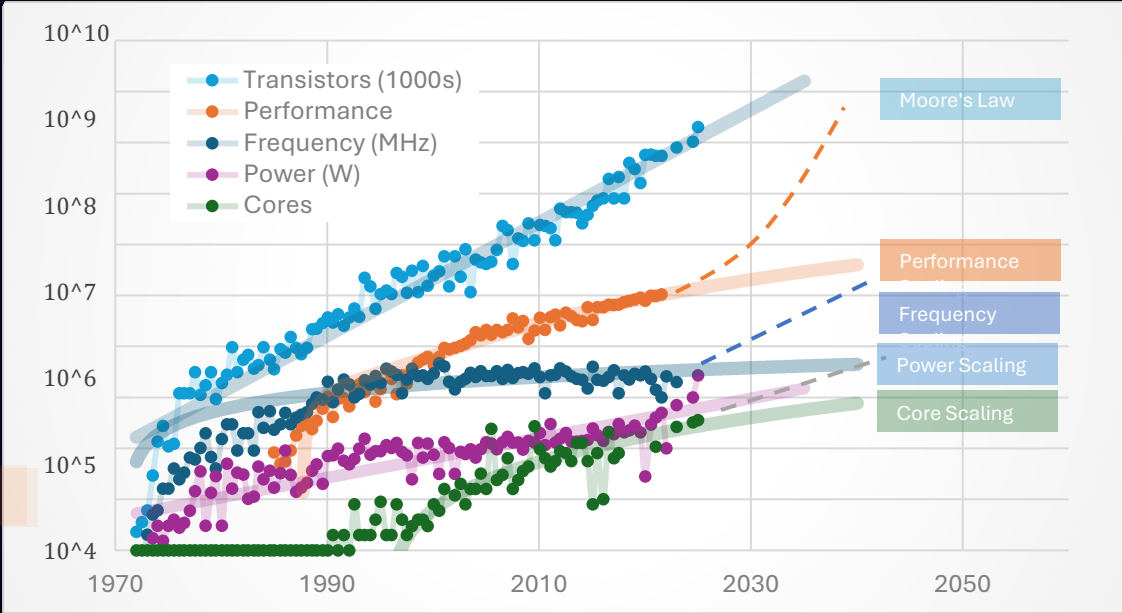
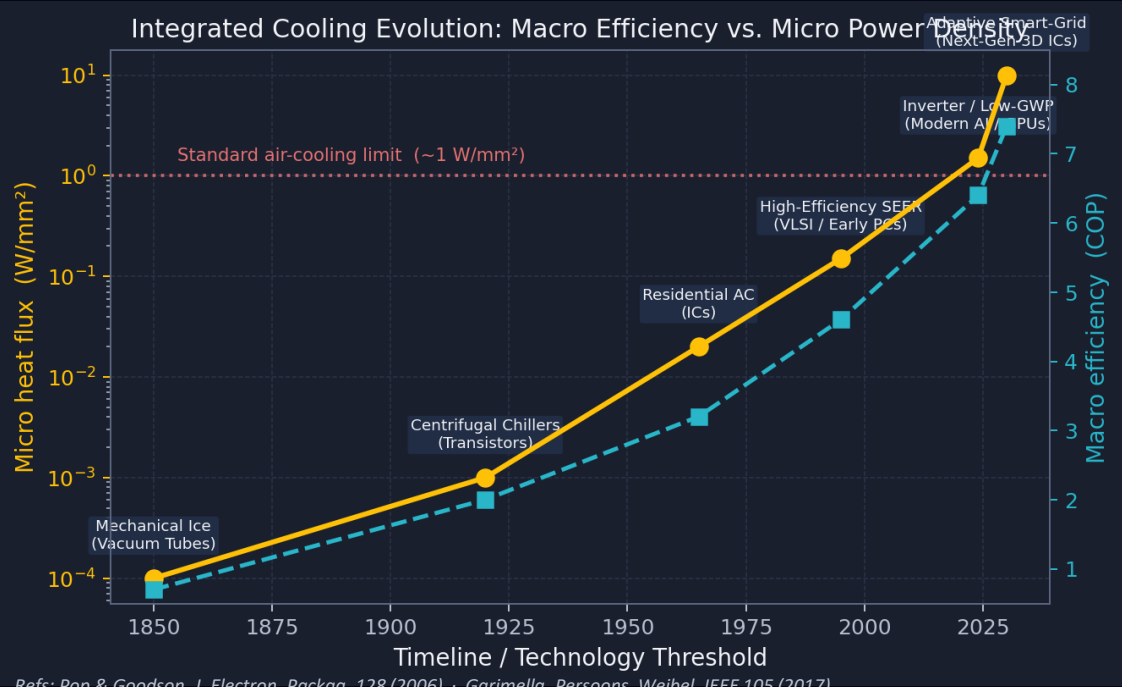
Macro (buildings, data centers): optimized for total energy efficiency — chillers, vapor cycles. COP rose from ~1 to ~7 over a century.

Micro (silicon): optimized for heat-flux removal and minimizing thermal resistance. Surface power density grew five orders of magnitude.

The next decade closes the gap:

micro-cooling must inherit macro's thermodynamic discipline (COP, exergy) without giving up flux density.

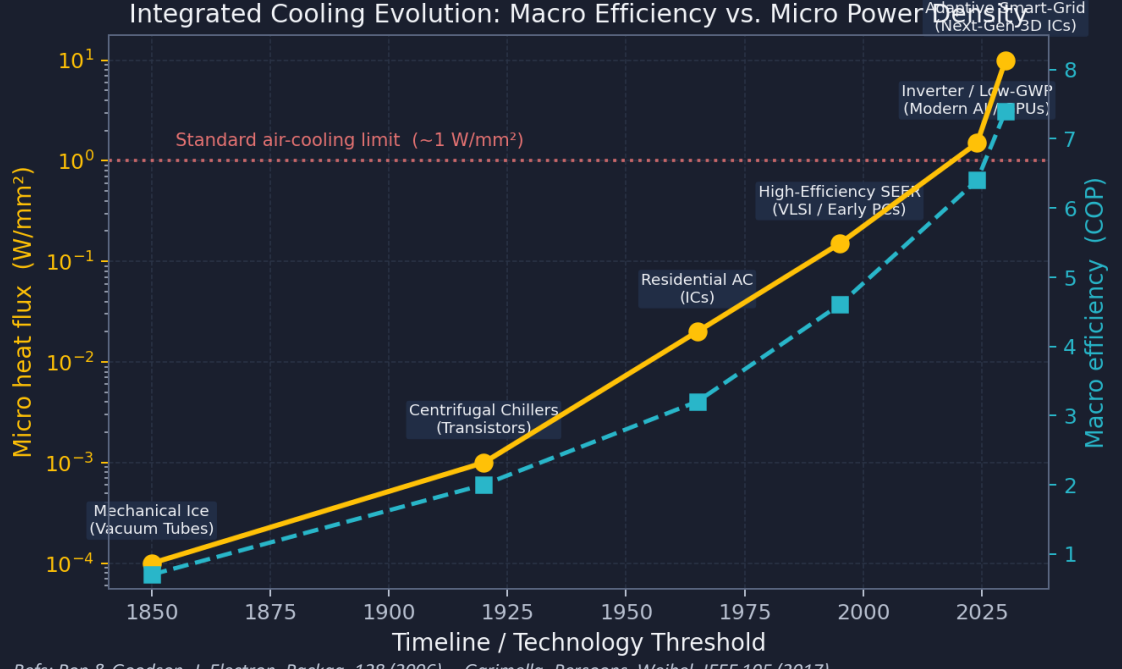
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2025+	Future High-Density SiC/GaN	5.0 – 10.0+	Micro-channels / Jet impingement



Evolution of Cooling Systems

The evolution of cooling systems follows **two distinct paths** : **Macro-scale systems**(buildings/ industry) optimizing total energy efficiency (COP), and **Micro-scale systems**(electronics/ chips) prioritizing *heat flux removal* and minimizing thermal resistance.

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🔥 POWER WALL

Conventional cooling fails above **1 W/mm²** —fans and heat sinks enter a cubic power regime.

💧 MICROFLUIDICS CEILING

Best microfluidic systems achieve **up to 10 W/mm²** —still far from the solar surface reference.

⚙️ SOLAR SURFACE REFERENCE

The solar surface radiates **63 W/mm²** —our ultimate physical cooling benchmark.

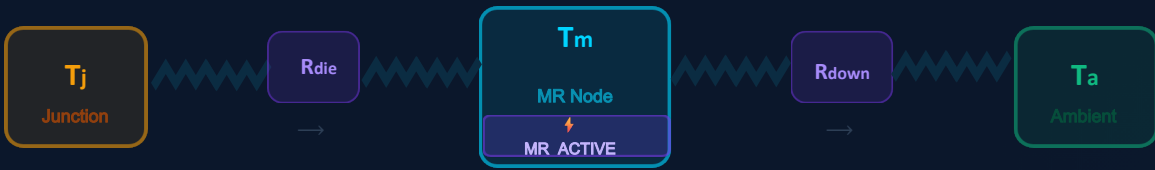
1) One slide on the difference between convection and microrefrigeration

2) What does microrefrigeration buy you? $R_{\text{die}} \rightarrow R_{\text{micro}} \rightarrow \text{etc...}$

The Simple '1D' Picture —Microrefrigeration Enables Zero-Resistance Die

Model the chip as a **series resistance stack** from junction T_j to ambient T_a . A microrefrigerator is inserted at internal node **T_m** , creating an **active thermal shunt**. **Goal:** minimize cooling work while enforcing $T_j \leq T_{lim}$. The sole control variable is **s** (MR strength).

1D THERMAL RESISTANCE STACK



$R_{down} = R_{TIM} + R_{ext}$ (TIM + cold-plate / convection)

MR STRENGTH

$s = q_m / q$
Fraction of chip flux removed by MR

DOWNSTREAM FLUX

$q_{down} = (1-s)q$
Flux continuing to bulk cooling path

GOVERNING EQUATIONS

Downstream resistance:

$$R_{down} = R_{TIM} + R_{ext}$$

Passive baseline (no MR, $s = 0$):

$$T_j - T_a = q(R_{die} + R_{down})$$

With MR active —effective resistance:

$$T_j - T_a = qR_{die} + (1-s)qR_{down}$$

Effective resistance as function of s :

$$R_{eff}(s) = R_{die} + (1-s)R_{down}$$

↑ Increasing s directly reduces apparent thermal resistance

⦿ OPTIMIZATION GOAL

Minimize total cooling work while enforcing $T_j \leq T_{lim}$. Bulk operating point is fixed. Only **s** is optimized.

Re-do: Vertical picture of real chip
and next to it the stack diagram

1. Physical problem

We consider a 1D thermal stack:

$$T_j \rightarrow R_{\text{die}} \rightarrow T_m \rightarrow R_{\text{down}} \rightarrow T_a.$$

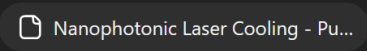
Here:

$$R_{\text{down}} = R_{\text{TIM}} + R_{\text{ext}}.$$

The passive chip temperature is

$$T_j - T_a = q(R_{\text{die}} + R_{\text{down}}).$$

This is the usual series thermal-resistance picture: heat must conduct through the die, then through TIM/cold plate/convection before reaching the ambient or coolant. The project background describes this as the micro-scale “conduction bottleneck,” where chip cooling is governed by heat-flux removal and thermal resistance rather than only macroscopic COP.



We now insert a microrefrigerator at the internal node T_m . It extracts heat before that heat crosses the downstream resistance. This creates an active thermal shunt.

The goal is:

$$\text{minimize cooling work while enforcing } T_j \leq T_{\text{lim}}.$$

In this simplified model:

$$\text{the bulk cooling operating point is fixed.}$$

We are **not** co-optimizing bulk flow, fan speed, or R_{ext} . The only control variable is the microrefrigeration strength s .

1. 1D fixed-bulk model

We model the chip as

$$T_j \rightarrow R_{\text{die}} \rightarrow T_m \rightarrow R_{\text{down}} \rightarrow T_a,$$

with

$$R_{\text{down}} = R_{\text{TIM}} + R_{\text{ext}}.$$

The microrefrigerator extracts heat at T_m . Define

$$s = \frac{q_m}{q},$$

so the heat continuing downstream is

$$q_{\text{down}} = (1 - s)q.$$

The bulk system is fixed; we are **not** co-optimizing flow, R_{ext} , or bulk COP.

2. Variables and definitions

Symbol	Meaning
q	chip heat flux
q_m	heat flux removed by the microrefrigerator
$s = q_m/q$	dimensionless microrefrigeration strength
T_j	junction / active-layer temperature
T_m	internal microrefrigerator node temperature
T_a	ambient or coolant reference temperature
T_{lim}	maximum allowable junction temperature
$\Delta T_{\text{lim}} = T_{\text{lim}} - T_a$	allowed junction temperature rise
R_{die}	upstream die resistance from junction to MR node
R_{down}	downstream resistance from MR node to ambient
R_{TIM}	thermal interface resistance
R_{ext}	external cold-plate/convection resistance
COP_m	microrefrigerator COP
COP_b	bulk cooling COP at fixed operating point
$COP_{b,0}$	chosen bulk reference COP
$\lambda_m = COP_m/COP_{b,0}$	normalized microrefrigerator COP
$\lambda_b = COP_b/COP_{b,0}$	normalized fixed-bulk COP

2. Thermal equations

$T_m - T_a = (1 - s)qR_{\text{down}}$

$T_j - T_m = qR_{\text{die}}$

so

$T_j - T_a = qR_{\text{die}} + (1 - s)qR_{\text{down}}$

or

$T_j - T_a = qR_{\text{eff}}(s)$

with

$R_{\text{eff}}(s) = R_{\text{die}} + (1 - s)R_{\text{down}}$

Thus s directly tunes the apparent thermal resistance.

3. Regimes of s

Define

$$s_0 = 1 + \frac{R_{\text{die}}}{R_{\text{down}}}.$$

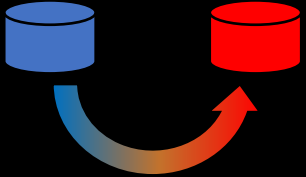
s range	Effective resistance	Node temperature	Physical meaning
$s = 0$	$R_{\text{die}} + R_{\text{down}}$	$T_m > T_a$	passive thermal stack
$0 < s < 1$	$R_{\text{die}} + (1 - s)R_{\text{down}}$	$T_m > T_a$	partial downstream cancellation
$s = 1$	R_{die}	$T_m = T_a$	full downstream cancellation
$1 < s < s_0$	$0 < R_{\text{eff}} < R_{\text{die}}$	$T_m < T_a$	partial die-resistance cancellation
$s = s_0$	0	$T_m = T_a - qR_{\text{die}}$	full resistance cancellation, $T_j = T_a$
$s > s_0$	< 0	$T_m < T_a - qR_{\text{die}}$	negative effective resistance, $T_j < T_a$

This table captures the high-level physical story: microrefrigeration first cancels the downstream path, then drives the internal node sub-ambient and cancels the die drop.

Convective Cold Plate Definitions

T_{in} : chiller-set inlet temperature

$$T_{out} = T_{in} + \frac{P_{heat}}{\rho C_p Q}$$



P_{heat} : Heat load entering on CP



$T_f = \beta(T_{in} + T_{out})$: Representative/effective fluid temperature inside CP ($\beta \approx 1/2$)

Quantity	Equation	Definition
Cold-plate thermal resistance	$R_{total}(Q, T_f) = R_{cp} + R_{bulk}$ $R_{cp} = R_{\infty} + \frac{R_{\infty} - R_0}{1 + \left(\frac{Q}{Q_c}\right)^m}$ $R_{bulk} = R_{passive} + \frac{B(T_f)}{Q}$	Q_c: Characteristic/design-point hydraulic roll-over point determined by microchannel/cold-plate characteristics point (boundary-layer thinning, turbulence onset, fin/channel utilization, pressure-drop rollover, Reynolds number) m: thermal roll-over exponent $\sim 0.5-1$ (air) or $\sim 0.7-1.5$ (liquid) $B(T_f) \sim \frac{\beta}{\rho C_p(T_f)}$ $R_{bulk} = 0$ for an air cold plate R_{∞}: Residual/flow-independent TIM/conduction/contact/spreading resistance R_0: Zero-flow or low-flow heat-transfer resistance
Pump/fan electrical power	$P_{pump}(Q) = P_{static} + \frac{\Delta p(Q/Q_c, T_f)Q}{\eta_{pump}(Q/Q_c, T_f)}$ $\approx P_{static} + \Phi(Q/Q_c)$ $\Phi(x) = P_c x^n \left[\frac{1+x^2}{2x} \right]^r$	- Hydraulic pressure $\Delta p(Q, T_f)Q \sim Q^\alpha$, $\alpha \approx 2(3)$ under laminal (turbulent) flow - Nominal fan/pump wall-plug efficiency curve $\eta_{pump}(x, T_f) \sim \eta_{max} \left[\frac{2x}{1+x^2} \right]^r$ $r \sim 0.5-1$: efficiency roll-off exponent - Peak efficiency $\eta_{max} \sim 0.1-0.5$ (air) $\sim 0.3-0.7$ (liquid) and peak efficiency operating point $Q_\eta \sim Q_c \Rightarrow x \sim 1$ (typically matched to hydraulic roll-over point for maximum efficiency/resistance tradeoff) P_c: Hydraulic power at Q_c P_{static}: Static motor/controller/bearing/electronics baseline (idle) power overhead
Chiller thermal load	$P_{chiller,th} = P_{heat} + \gamma P_{pump} + P_{leak}$	- Heat the chiller must remove from the cold loop. γ: Fraction of pump electrical power that heats the cold stream P_{leak}: Heat leak from ambient into chilled lines, manifolds, insulation, reservoir.
Chiller electrical power	$P_{chiller} = \frac{P_{chiller,th}(Q)}{COP_{chiller}(T_{in}, T_{amb})}$	Electrical power required by the chiller to maintain the fluid setpoint.
Total electrical power	$P_{total} = P_{pump} + P_{chiller}$	Total pump/fan plus chiller electrical power.

Thermal wall & COP: 1D model of hybrid convection + μ -refrigeration

Knobs/optimization parameters: flow rate $u = Q/Q_c$, refrigeration ratio $s = q_\mu/q$

Definitions:

$$R_{eff}(u, s) \equiv (1 - s)R_{ext}(u) + R_{die}$$

$$R_{ext}(u) = R_\infty + \frac{R_0 - R_\infty}{1 + u^m}$$

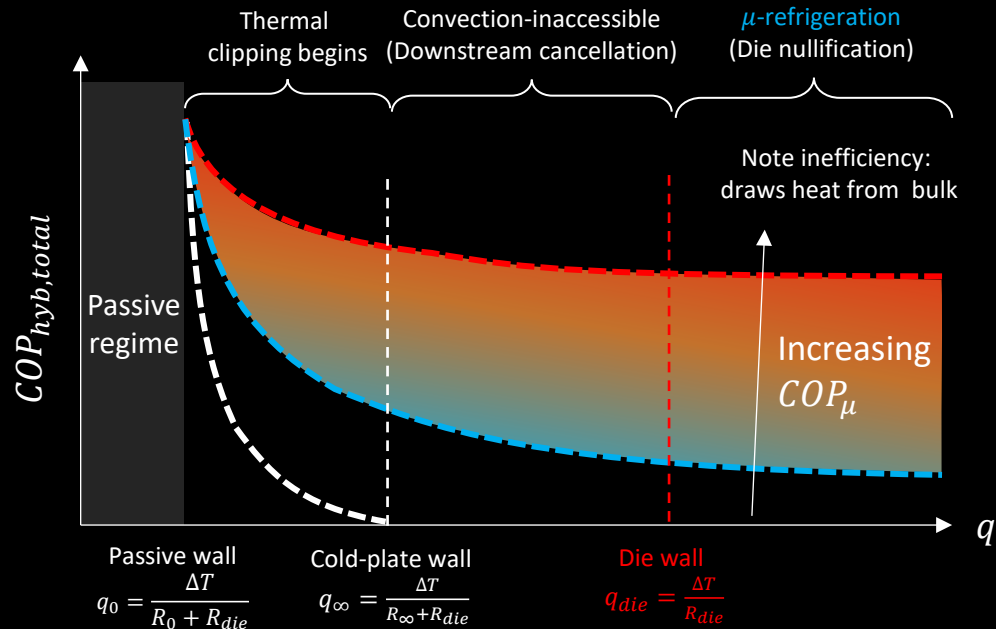
$$q_{0,\infty} = \frac{\Delta T}{R_{0,\infty} + R_{die}}$$

$$P_{pump}(u) = P_{static}^0 + P_c u^n \left[\frac{1 + u^2}{2u} \right]^r$$

$$q_{die} = \frac{\Delta T}{R_{die}}$$

$$P_{total} = \frac{P_{pump}(u)}{A} + s q / COP_\mu$$

$$\Phi(q) = \frac{\Delta T}{q} - R_{die}$$



Objective:

$$COP_{hyb,active}(u, s) \equiv \frac{q - q_0}{P_{total}(u, s)}$$

$$\text{s.t. } R_{eff}(u, s) \leq \frac{\Delta T}{q}$$

$$COP_{hyb,total} \equiv \frac{q}{P_{total}(u, s)} = \left(\frac{q}{q - q_0} \right) COP_{hyb,active}(u, s)$$

Solution:

$$COP_{hyb,active}(q) = \begin{cases} \infty, & q < q_0 \text{ (Passive)} \\ \frac{q - q_0}{P_{pump}(u_{opt}(q)) + \frac{q s_{opt}(q)}{COP_\mu}}, & q \geq q_0 \text{ (Active)} \end{cases}$$

$$s_{opt}(q) = \begin{cases} 0, & q < q_0 \\ 1 - \frac{\Phi(q)}{R_{ext}(u_{opt}(q))}, & q \geq q_0 \end{cases}$$

$$u_{opt}(q) = \begin{cases} 0, & q < q_0 \text{ (Passive)} \\ \min[u_{hyb}(q), u_{conv}(q)], & q_0 \leq q \leq q_\infty \text{ (Thermal clipping)} \\ u_{hyb}(q), & q_\infty \leq q \leq q_{die} \text{ (Downstream nullification)} \\ 0, & q \geq q_{die} \text{ (Die cancellation)} \end{cases}$$

$$u_{conv}(q) = u \quad \text{s.t.} \quad R_{ext}(u) = \Phi(q)$$

$$u_{hyb}(q) = u \quad \text{s.t.} \quad \frac{dP_{pump}(u)}{du} + \frac{q}{COP_\mu} \frac{d}{du} \left[1 - \frac{\Phi(q)}{R_{ext}(u)} \right] = 0$$

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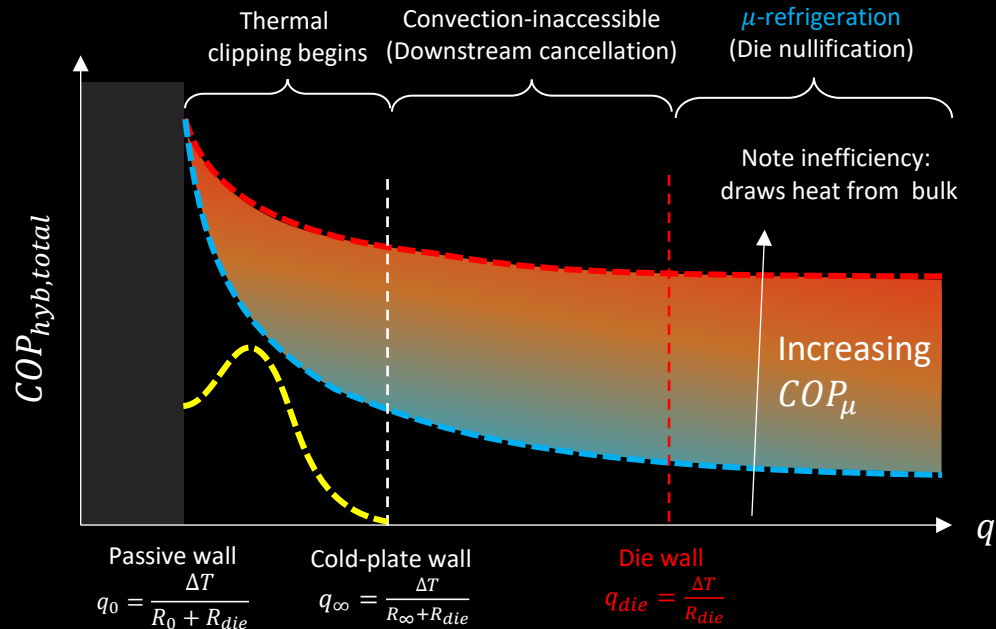
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When “normal” convective optimization breaks down, even sub-ambient cooling or low-COP micro-refrigeration can still make system-level sense. At extreme conditions, cooling changes:

- allowable power density,
- leakage power,
- device reliability,
- clock frequency,
- or package feasibility.

Thus, even $COP < 1$ can improve total system efficiency or enable operation that would otherwise be impossible.

1. Extremely high heat flux

At high fluxes, increasing flow becomes expensive while thermal benefits saturate. This is precisely the regime where:

- convection nears the cold plate wall
- localized MR can attack the hotspot directly.

Even if $COP_{\mu} < 1$, it may still reduce required flow or avoid impossible package temperatures.

Technology	Local heat flux
CPU/GPU	$\sim 1 \text{ W/mm}^2$
AI accelerators	$> 1 \text{ W/mm}^2$ hotspots
Laser/photonics	$> 10 \text{ W/mm}^2$
3D stacks	severe localized hotspots

2. Leakage-dominated operation

Leakage power scales $P_{leak} \sim e^{-eV_{th}/kT} \Rightarrow$ a modest temperature reduction can dramatically reduce chip power. For advanced nodes, a 10–20°C reduction can reduce leakage by 20–60%.

Hence, cooling has multiplicative benefit, reducing the heat generation itself. This requires that we extend our model to $q = q_{dyn} + q_{leak}(T_j)$

Node	Leakage impact
7 nm	substantial
5 nm	severe
3 nm	dominant

3. Near-package thermal limits

- solder fatigue
- TSV limits
- interposer delamination
- HBM thermal limits
- optical alignment drift
- electromigration
- thermal runaway

Even $COP_{\mu} \ll 1$ may still be optimal if it avoids:

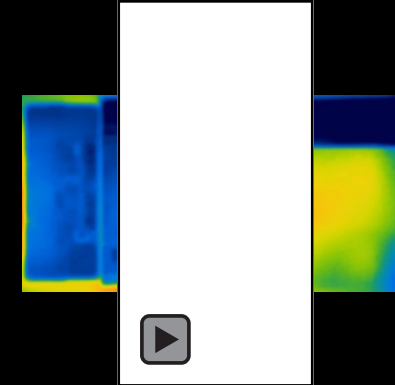
- throttling/latency/timing issues
- reduced yield
- lower clock speeds
- package redesign
- or total infeasibility.

4. Next up, how to model:

- hotspot clipping,
- DVFS expansion
- boost/throttle behavior
- leakage reduction
- transient stabilization
- reliability effects
- package/system interactions

What's missing? In real chips, spatial localization, nonlinear bulk costs, and co-design amplify this into a much larger system-level advantage.

- **Spatial selectivity & hot-spot physics:** Real chips are not uniform: small-area hot spots ($\alpha/\beta \gg 1$) dominate temperature limits. Microrefrigeration can target only these regions, delivering disproportionate benefits compared to uniform cooling.
- **Spreading + vertical resistance cancellation:** Beyond screening/canceling downstream resistance, localized cooling suppresses lateral spreading/constriction resistance—often the dominant penalty for dense hot spots—effectively “flattening” the thermal landscape.
- **Decoupled control of peak vs. average heat:** Instead of forcing one mechanism to handle both hot-spot and background refrigeration, microrefrigeration allows independent control of peak temperature and total heat load.
- **Co-design of the cooling stack:** Instead of layering microrefrigeration onto a fixed cooler, the bulk system can be redesigned around the reduced load—shifting operation back to its high-efficiency regime. Reducing residual heat to the cold plate allows throttling fans/pumps.
- **Optimal load sharing (“thermal gearbox”):** The hybrid system naturally splits work: microrefrigeration handles high-flux local peaks, while bulk cooling handles low-flux average heat efficiently—maximizing overall COP.
- **Architecture awareness & scalability:** Benefits grow with modern trends (CPUs, GPUs, AI accelerators, 3D stacks) where power is more localized and spreading penalties are larger.
- **Reduced thermal cross-talk and dynamic targeting:** Local cooling prevents hot units from heating neighbors and can be steered to active regions, aligning cooling effort with computation.



Beyond 1D: Hot Spots & the Geometric Constriction Crisis

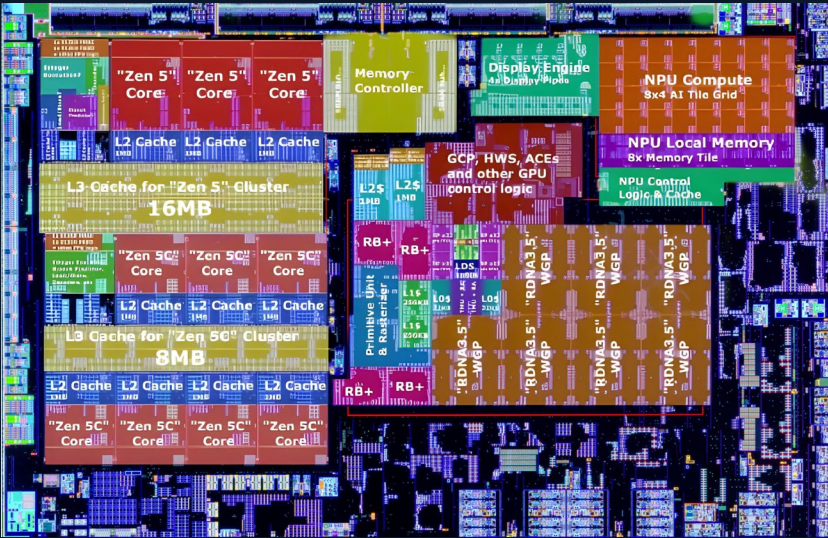
⚠ THE HIDDEN CRISIS

Even if our 1D model says **$q = 1 \text{ W/mm}^2$ is safe**, a local hot spot at **10 W/mm^2** instantly exceeds 100°C due to **constriction (spreading) resistance**—heat must 'bloom' from a tiny functional unit into a larger volume.

WHY 1D IS INSUFFICIENT

- ✂ **Constriction Resistance R_{sp} :** Heat from tiny functional unit must "bloom" into larger volume → massive local temperature spike not captured by 1D averages.
- 🔌 **Spatial Mismatch:** Bulk cooling extracts heat globally. A hot spot at 10 W/mm^2 surrounded by idle silicon is invisible to the bulk system.

MR Advantage: Microrefrigerators provide **"Gradient Flattening" + "Decoupling"** —surgical vs. blunt-instrument bulk cooling.



1D ASSUMPTION VS MODERN REALITY

FEATURE	1D ASSUMPTION	MODERN REALITY
Heat Path	Vertical only (R_{die})	Spreading resistance ($R_{die} + R_{sp}$)
Flux (q)	Uniform (q_{avg})	Non-Uniform ($q_{peak} \gg q_{avg}$)
Cooling	Global extraction	Spatial mismatch / Waste heat bleed
Thermal Limit	Set by bulk k	Set by local "Heat Spreading & conduction wall" limits

CONSTRICTION EFFECT: A/B AMPLIFICATION

Hot-Spot Flux Amplification

$$q_H = (\alpha/\beta) \cdot q$$

e.g. $\alpha/\beta = 5 \rightarrow$ hot spot carries $5\times$ chip-average flux

MR Opportunity Scales With α/β

Higher $\alpha/\beta \rightarrow$ bigger spreading resistance $\rightarrow$ larger thermal penalty $\rightarrow$ higher MR value per watt removed

R_{sp} is not captured in 1D model—the 2D model extends thermodynamics to account for spatial heat concentration

To analyze the advantage of **spatially-selective cooling**, we must move from a single-loop circuit to a **Multi-Node Network**. This allows us to see how "spot" cooling effectively manages the "System COP" by avoiding the energy waste of cooling the entire die to satisfy a single hot spot.

Summary: The "Surgical" Thermal Circuit		
Circuit Element	Targeted Micro-Refrigerator	Bulk Convection / Chiller
Topology	Shunt Regulator at the Hot Spot.	Series Regulator for the whole loop.
Waste Heat	Localized; recycled via LPC	High; must be handled by the entire loop.
System COP	High. Power is only spent where the bottleneck is "pressing."	Low. Energy is wasted cooling "dark silicon" or idle cores.
Performance	Allows "Overclocking" of the hot core without global throttling.	Limited by the "average" ability of the heatsink.

Hot spot relief: 2d model of hybrid convection + μ -refrigeration

Knobs/optimization parameters:

flow rate $u = Q/Q_c$ refrigeration ratio $s = q_\mu/q$

Total chip power $Q = q_H A_H + q_B A_B$

Total chip area $A = A_H + A_B$

Average chip flux $\bar{q} = \frac{Q}{A}$

Hot spot power fraction $\alpha = \frac{q_H A_H}{Q}$

Hot spot area fraction $\beta = \frac{A_H}{A}$

Hot spot thermal pressure $\gamma_H = \frac{q_H}{\bar{q}} = \frac{\alpha}{\beta}$

Bulk thermal pressure $\gamma_B = \frac{q_B}{\bar{q}} = \frac{1-\alpha}{1-\beta}$

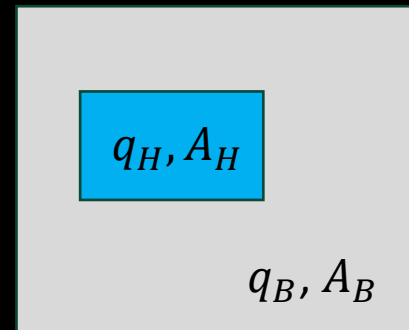
2D thermal stack model:

$$\begin{pmatrix} \Delta T_H \\ \Delta T_B \end{pmatrix} = \mathbf{R}^\uparrow(u) \begin{pmatrix} \gamma_H(1-s) \\ \gamma_B \end{pmatrix} \bar{q} + \mathbf{R}^\downarrow \begin{pmatrix} \gamma_H \\ \gamma_B \end{pmatrix} \bar{q}$$

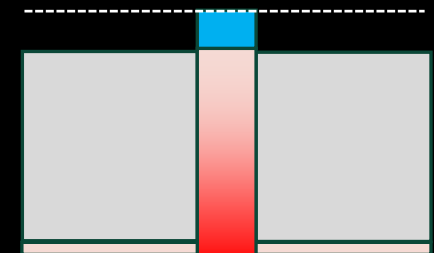
$$= \begin{pmatrix} R_{HH}^\uparrow & R_{HB}^\uparrow \\ R_{BH}^\uparrow & R_{BB}^\uparrow \end{pmatrix} \begin{pmatrix} \gamma_H(1-s) \\ \gamma_B \end{pmatrix} \bar{q} + \begin{pmatrix} R_{HH}^\downarrow & R_{HB}^\downarrow \\ R_{BH}^\downarrow & R_{BB}^\downarrow \end{pmatrix} \begin{pmatrix} \gamma_H \\ \gamma_B \end{pmatrix} \bar{q}$$

$$\Delta T_H = \bar{q} [\gamma_H R_{HH}^\downarrow + \gamma_B R_{HB}^\downarrow + \gamma_H(1-s) R_{HH}^\uparrow(u) + \gamma_B R_{HB}^\uparrow(u)]$$

$$\Delta T_B = \bar{q} [\gamma_H R_{BH}^\downarrow + \gamma_B R_{BB}^\downarrow + \gamma_H(1-s) R_{BH}^\uparrow(u) + \gamma_B R_{BB}^\uparrow(u)]$$



$$\mathbf{R}^\uparrow(u) = R_{ext}(u) \mathbf{I} + \xi \mathbf{R}_{sp}$$



$$\mathbf{R}^\downarrow = R_{die} \mathbf{I} + (1 - \xi) \mathbf{R}_{sp}$$

External/cold-plate resistance

$$R_{ext}(u) = R_\infty + \frac{R_0 - R_\infty}{1 + u^m}$$

Spreading resistance matrix:

$$\mathbf{R}_{sp} = \begin{pmatrix} R_{sp,H} & R_c \\ R_c & R_{sp,B} \end{pmatrix}$$

Constriction & spreading resistance

$$R_c = \rho_c \sqrt{R_{sp,H}(\beta) R_{sp,B}(\beta)}$$

$$R_{sp,H}(\beta) = \chi(\beta^{-p} - 1)$$

$$R_{sp,B}(\beta) = \chi[(1 - \beta)^{-p} - 1]$$

Typical parameters:

$$\begin{aligned} p &\cong 0.5, \\ \chi &\cong 0.001, \\ \rho_c &\cong 0.5, \\ \xi &\cong 0.5 \end{aligned}$$

2. Heat partition and hot-spot geometry

Let

$$q = \frac{Q}{A}$$

be the average chip heat flux, where Q is total chip power and A is chip area.

Let a hot-spot area fraction β dissipate a power fraction α . Then the local hot-spot heat flux is

$$q_H = \frac{\alpha}{\beta} q.$$

The background/rest-of-chip heat flux is

$$q_C = \frac{1 - \alpha}{1 - \beta} q.$$

The ratio

$$\frac{\alpha}{\beta}$$

is the hot-spot flux amplification. A value $\alpha/\beta = 5$, for example, means the hot spot carries five times the chip-average heat flux.

3. Core variable table		
Symbol	Type	Meaning
q	operating variable	average chip heat flux
Q	operating variable	total chip power
A	design variable	chip area
T_a	environment	ambient/coolant reference temperature
T_{lim}	constraint	maximum allowed hot-spot temperature
$\Delta T_{\text{lim}} = T_{\text{lim}} - T_a$	constraint	allowed hot-spot temperature rise
α	workload/layout parameter	fraction of total chip power in hot spot
β	layout parameter	fraction of chip area occupied by hot spot
$q_H = (\alpha/\beta)q$	derived	hot-spot heat flux
$q_C = ((1 - \alpha)/(1 - \beta))q$	derived	background heat flux
n	bulk model	heat-transfer exponent, $h \sim g^n$
m	bulk model	pump/fan exponent, $W_{\text{pump}} \sim g^m$
ϕ_0	bulk model	fixed/idle overhead fraction for fixed installed hardware
q_0	reference	fixed-hardware design heat flux
R_{ref}	reference	co-designed bulk-loop reference resistance
γ	co-design exponent	resistance-effort exponent, roughly $\gamma \sim m/n$

R_v	stack design	vertical die resistance above microrefrigerator location
$R_{\text{sp}}(\beta)$	thermal geometry	hot-spot spreading/constriction resistance
$R_m = t_m/k_m$	MR design	passive resistance of inserted microrefrigerator layer
R_{TIM}	package design	thermal interface resistance below the MR layer
R_{ext}	bulk cooling	external cold-plate/convection resistance
g	Option A control	normalized flow/fan/pump drive
s_H	control	hot-spot-selective microrefrigeration fraction
s_C	control	background/rest-of-chip microrefrigeration fraction
M	derived	total fraction of chip heat removed by microrefrigeration
λ_H	technology	$COP_{m,H}/COP_{b,0}$
λ_C	technology	$COP_{m,C}/COP_{b,0}$
η_{sp}	design/effectiveness	fraction of spreading resistance cancelled by local MR
ξ_m	MR geometry	fraction of R_m downstream of MR cooling centroid



Hot spot relief: 2d model of hybrid convection + μ -refrigeration

Knobs/optimization parameters:

flow rate $u = Q/Q_c$ refrigeration ratio $s = q_\mu/q$

Definitions:

$$R_{eff,H}(u, s) = \gamma_H R_{HH}^\downarrow + \gamma_B R_{HB}^\downarrow + \gamma_H(1-s)R_{HH}^\uparrow(u) + \gamma_B R_{HB}^\uparrow(u)$$

$$R_{eff,B}(u, s) = \gamma_H R_{BH}^\downarrow + \gamma_B R_{BB}^\downarrow + \gamma_H(1-s)R_{BH}^\uparrow(u) + \gamma_B R_{BB}^\uparrow(u)$$

$$\Phi(q) = \frac{\Delta T}{q} - \gamma_H R_{HH}^\downarrow - \gamma_B R_{HB}^\downarrow$$

Note: $R_{eff,B}(u, s) \leq \frac{\Delta T}{q}$ is currently not enforced here; this is problematic if $q > q_{\bar{u},B}$ (bulk cooling requires active cooling), which would preclude $u_{opt} \rightarrow 0$

$$\bar{q}_{die,H} = \frac{\Delta T}{\gamma_H R_{HH}^\downarrow + \gamma_B R_{HB}^\downarrow}$$

$$\bar{q}_{\bar{u} \in \{0, \infty\}, H} = \frac{\Delta T}{\gamma_H R_{HH}^\downarrow + \gamma_B R_{HB}^\downarrow + \gamma_H R_{HH}^\uparrow(\bar{u}) + \gamma_B R_{HB}^\uparrow(\bar{u})}$$

$$P_{total} = \frac{P_{pump}(u)}{A} + \alpha s q / COP_\mu$$

$$P_{pump}(u) = P_{static} + P_c u^n \left[\frac{1+u^2}{2u} \right]^r$$

Note: $\bar{q}_{\bar{u} \in \{0, \infty\}, H}$, q_{die} are the average-chip fluxes at the corresponding hot-spot walls; the actual/corresponding hot-spot wall $q_{H,\bar{u}} = \gamma_H q_{\bar{u},H}$

Objective:

$$COP_{hyb,active}(u, s) \equiv \frac{\bar{q} - \bar{q}_{0,H}}{P_{total}(u, s)}$$

$$\text{s.t. } R_{eff,H}(u, s) \leq \frac{\Delta T}{q}$$

Solution:

$$COP_{hyb,total} \equiv \frac{\bar{q}}{P_{total}(u, s)} = \left(\frac{\bar{q}}{\bar{q} - \bar{q}_{0,H}} \right) COP_{hyb,active}(u, s)$$

$$COP_{hyb,active}(q) = \begin{cases} \infty, & q < q_0 \text{ (Passive)} \\ \frac{\bar{q} - \bar{q}_{0,H}}{P_{pump}(u_{opt}(q)) + \frac{\alpha q s_{opt}(q)}{COP_\mu}}, & q \geq q_0 \text{ (Active)} \end{cases}$$

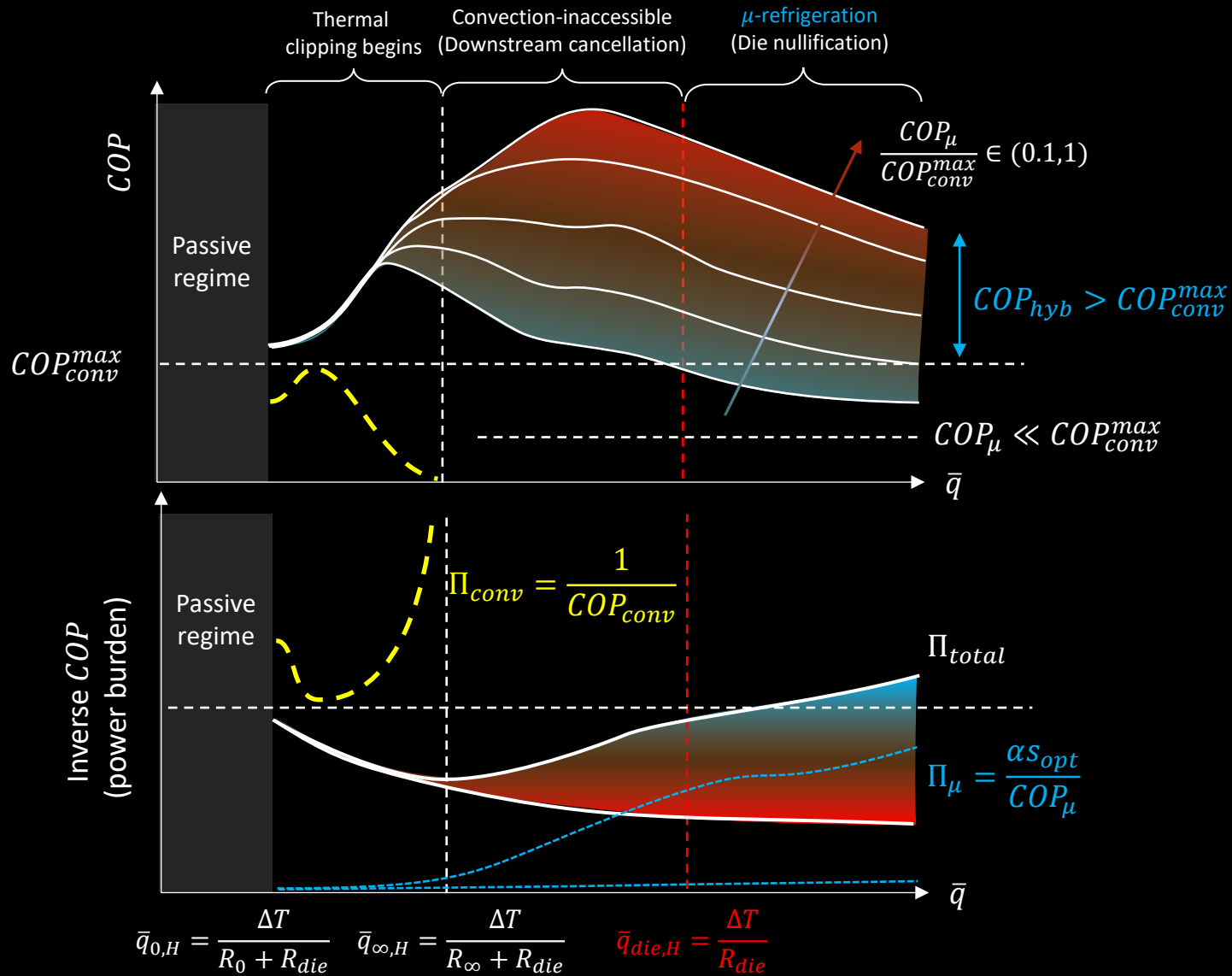
$$s_{opt}(q) = \begin{cases} 0, & \bar{q} < \bar{q}_{0,H} \\ 1 - \frac{\Phi(\bar{q}) - \gamma_B R_{HB}^\uparrow(u_{opt}(q))}{\gamma_H R_{HH}^\uparrow(u_{opt}(q))}, & \bar{q} \geq \bar{q}_{0,H} \end{cases}$$

$$u_{opt}(q) = \begin{cases} 0, & \bar{q} < \bar{q}_{0,H} \text{ (Passive)} \\ \min[u_{hyb}(\bar{q}), u_{conv}(\bar{q})], & \bar{q}_{0,H} \leq \bar{q} \leq \bar{q}_{\infty,H} \text{ (Thermal clipping)} \\ u_{hyb}(\bar{q}), & \bar{q}_{\infty,H} \leq \bar{q} \leq \bar{q}_{die,H} \text{ (Upstream nullification)} \\ 0, & \bar{q} \geq \bar{q}_{die,H} \text{ (Die cancellation)} \end{cases}$$

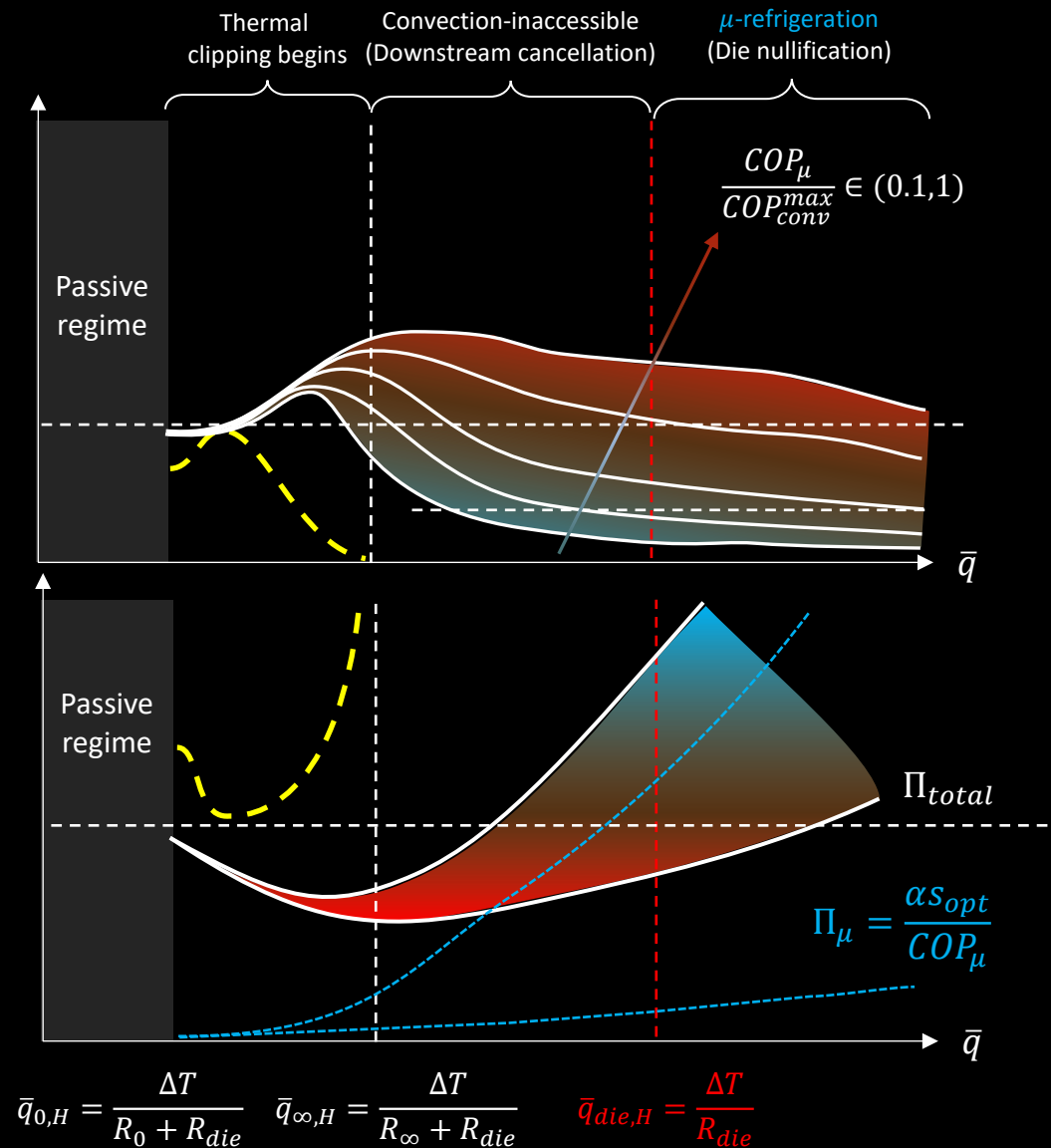
$$u_{conv}(q) = u \quad \text{s.t.} \quad \gamma_H R_{HH}^\uparrow(u) + \gamma_B R_{HB}^\uparrow(u) = \Phi(q)$$

$$u_{hyb}(q) = u \quad \text{s.t.} \quad \frac{dP_{pump}(u)}{du} + \frac{\alpha q}{COP_\mu} \frac{d}{du} \left[1 - \frac{\Phi(q) - \gamma_B R_{HB}^\uparrow(u)}{\gamma_H R_{HH}^\uparrow(u)} \right] = 0$$

Strong utility : $\Pi_{Pump} + \frac{\alpha_{Sopt}}{COP_{\mu}} \lesssim \Pi_{conv}^{max}$



Weak utility : $\Pi_{Pump} + \frac{\alpha_{Sopt}}{COP_{\mu}} \gg \Pi_{conv}^{max}$



When does thermal clipping (hot spot relief) = higher system COP?

The most compact interpretation is:

$$\Lambda \equiv \frac{COP_m}{\alpha s}$$

$$s = 1 - \frac{\Phi(q) - \gamma_B R_{HB}^\dagger(u_{opt}(q))}{\gamma_H R_{HH}^\dagger(u_{opt}(q))}$$

- $\Lambda \gtrsim 1$: MR is system-useful.
- $\Lambda \ll 1$: MR is energetically expensive.

Regime	Geometry / power distribution	Key discriminator	Thermal situation
1. Useful low-COP MR	Small hot spot area and small hot-spot power fraction $\beta \ll 1$, $\alpha \ll 1$, $\gamma_H = \alpha/\beta \gg 1$	$\alpha s \lesssim COP_m$	Convection handles bulk chip power, but hot spot hits thermal wall first
2. Expensive MR / bulk-like refrigeration	Large hot-spot power fraction and/or broad hot region $\alpha = O(0.1-1)$ and/or $\beta \not\ll 1$	$\alpha s \gg COP_m$	MR effectively cools a large fraction of the chip, or requires strong overcooling $s \gg 1$

Spatial Scaling of Power Density

Analysis of the 5-order of magnitude growth from rack to hot spot

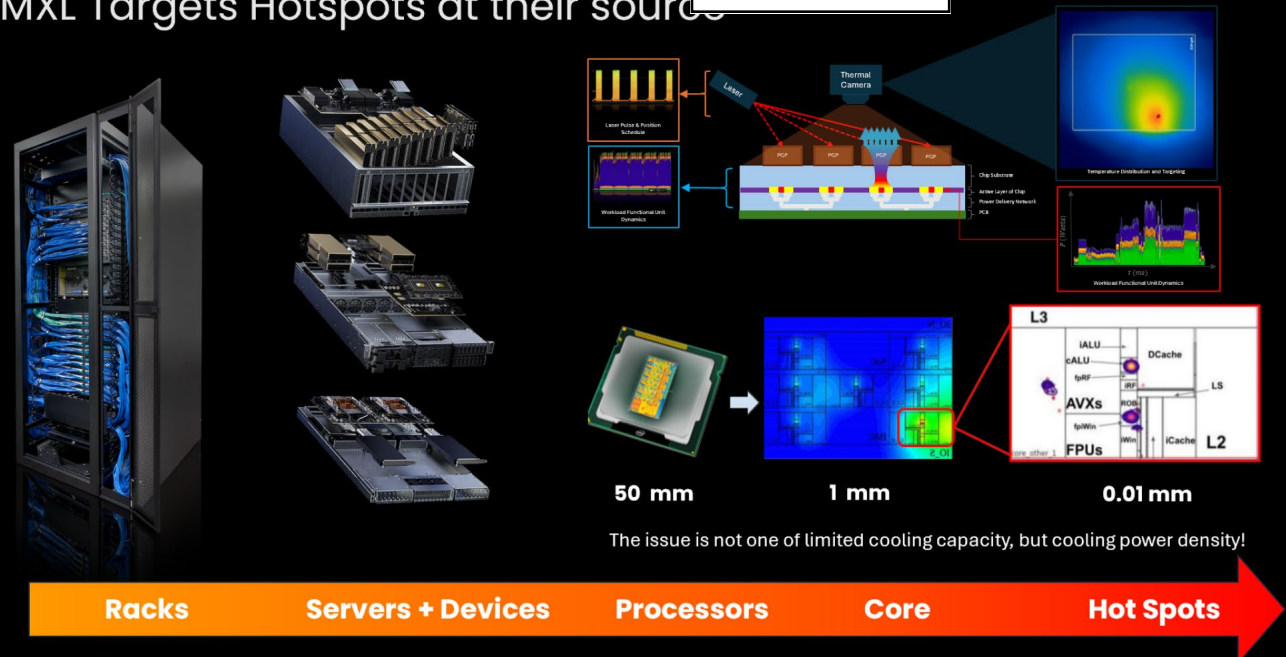
*"The issue is not limited cooling capacity—it is **cooling power density**."*

	Racks:	0.01 W/mm ²
	Servers:	0.1 W/mm ²
	Processors:	1.0 W/mm ²
	Cores:	10.0 W/mm ²
	Hot Spots:	100 W/mm ²

HEAT FLUX SCALING



Heat Starts Small But Scales Massive
MXL Targets Hotspots at their source



Is it a one-size-fits all?

No! Different regimes and architectures of competency (what the trends suggest)

"The value of LAMBDA defines our strategic pivot. In this ASF-cooled architecture, LAMBDA represents a stable efficiency floor. By monitoring the marginal collapse of our fans, we engage the 'Zero-Resistance' limit at the exact point of efficiency parity. This ensures that the hot spot is surgically removed from the thermal circuit, allowing the bulk system to stay in its high-efficiency regime while the micro-refrigerator handles the performance scaling."

Chip type	Thermal signature	COP impact	Performance / latency impact	Reliability / failure impact	TDP / packaging impact	Microrefrigeration opportunity
General-purpose CPU	Bursty localized hot spots in cores, caches, AVX units	Bulk cooler overreacts to short local peaks	Boost clocks drop; latency jitter during thermal excursions	Repeated thermal cycling stresses bumps/TIM	TDP limited by worst-case boost region	Moderate–high: targeted cooling can preserve boost without overcooling whole die
GPU / AI accelerator	Dense sustained hot regions across SMs/tensor cores	Fans/pumps pushed toward inefficient high-flow regime	Sustained throttling lowers TFLOPs and training throughput; liquid-cooled H100 systems have shown higher thermal stability and performance than air-cooled ones	Higher temperature increases leakage and accelerates aging	Package and rack cooling become limiting before silicon compute does	Very high: local MR can shave compute hot spots while bulk handles average heat
GPU + HBM package	Compute die and memory stacks thermally coupled	Cooling must satisfy both GPU and HBM temperature limits	Memory bandwidth, timing margin, and GPU clocks become coupled	HBM stacks trap heat and amplify thermomechanical stress; conventional mitigations reduce but do not eliminate internal resistance (MDPI)	HBM placement, interposer, and cold-plate path constrain TDP	Very high: MR can decouple GPU hot spots from HBM temperature limits
3D-stacked logic / chiplets	Heat trapped in buried layers; weak vertical escape	Bulk COP collapses because external cooling cannot reach internal hot spots effectively	Severe throttling from buried junctions; poor thermal observability	Hot buried layers increase electromigration, bonding stress, and fatigue	Stack height and bonding density limited by heat removal	Extremely high: embedded/local MR can create an internal heat-exit path
HBM / DRAM stacks	Thermally sensitive memory cells; vertical heat trapping	Cooling energy spent to protect retention margins	Higher refresh rates, timing guardbands, and latency penalties	Temperature worsens leakage/retention and long-term reliability; hot-spot reduction can reduce leakage significantly in modeled AI-chip stacks (Springer)	Limits stack height and bandwidth density	High: MR can stabilize memory temperature without globally chilling compute
Mobile SoC / edge AI	Small package, low thermal mass, skin-temperature limits	Passive spreading dominates; little fan/pump leverage	Fast thermal throttling; user-perceived latency spikes	Battery and package temperature constraints dominate	TDP capped by enclosure and comfort limits	Moderate: useful for burst smoothing and local peak shaving
RF / GaN / SiC power electronics	Very high local heat flux, small active regions	Bulk cooling inefficient because heat is extremely localized	Gain, efficiency, and switching performance degrade with temperature	Hot spots accelerate defects, electromigration, and dielectric failure	Power density limited by junction thermal path	Very high: MR directly attacks the local flux bottleneck
Data-center accelerator module	Many chips share rack/cold-plate constraints	One hot device can force high coolant flow for the whole node	Cluster-level throttling, scheduling inefficiency, reduced throughput	Temperature-correlated failures and service events increase	Rack power/cooling provisioning becomes conservative	High: MR can reduce peak device demand and keep shared cooling in efficient regime

Rank	Chip type	β : hotspot area / die area	α : hotspot power / total power	Local amplification (thermal pressure) α/β	Typical hotspot flux	Thermal pressure interpretation
1	NAND / low-power memory	0.2–0.6	0.2–0.6	~0.3–3	0.01–0.1 W/mm ²	Low absolute flux and weak amplification; bulk/passive cooling usually sufficient.
2	DRAM / commodity memory	0.1–0.4	0.2–0.5	~0.5–5	0.05–0.3 W/mm ²	Mild local pressure; hotspots exist but rarely dominate system COP.
3	HBM stack	0.05–0.25	0.2–0.6	~0.8–12	0.2–1 W/mm ²	Moderate-to-high pressure; vertical stack path matters as much as local flux.
4	Mobile SoC	0.02–0.15	0.2–0.5	~1.3–25	0.5–3 W/mm ²	Bursty high pressure; spatial/dynamic spot cooling becomes valuable.
5	Server CPU	0.02–0.10	0.2–0.5	~2–25	1–8 W/mm ²	High pressure; boost clocks are limited by local sensors, not average die temperature.
6	GPU / AI accelerator	0.05–0.20	0.3–0.7	~1.5–14	1–10 W/mm ²	Broad but intense pressure; hybrid cooling useful because power is high and persistent.
7	Chiplet / 2.5D HPC package	0.01–0.10	0.2–0.6	~2–60	2–15 W/mm ²	Very high pressure from package nonuniformity; bulk cooling sees the average but performance sees the peak.
8	3D logic-on-logic / stacked compute	0.005–0.05	0.2–0.6	~4–120	5–30 W/mm ²	Severe pressure; embedded or optical microrefrigeration becomes a near-mandatory bypass.
9	RF / GaN / SiC power IC	0.001–0.03	0.2–0.8	~7–800	20–300+ W/mm ²	Extreme pressure; local amplification and absolute flux both exceed what bulk convection can efficiently address.

$$\lambda = \frac{COP_{\mu}}{COP_{conv}}$$

(1) Here is the **Decision Framework** classifying different chip architectures by their thermal signatures (α , β) and the resulting cooling strategy.

Architectural Decision Framework: Cooling Strategy by Chip Type

Architecture	Signature (α , β)	Thermal Profile	Optimal Hybrid Strategy
Standard CPU	Low α , High β	Balanced/Uniform	Bulk-Dominant: R_{ext} is the main lever. Active cooling is only for peak bursts.
GPU / AI	High α , Mid β	Dense Parallel Loads	Hybrid Steady-State: Active extraction handles the compute cores; Bulk handles HBM/Memory.
AI Accelerator	High α , Low β	"Needle" Hot Spots	Surgical Dominant: Micro-refrigeration is mandatory to prevent instant throttling of functional units.
3D Stacked IC	Ultra-High α/β	Volumetric Heat	Embedded Active: Traditional convection cannot reach inner layers; "Negative Resistance" is the only path.

The Integrated Decision Map: λ vs. Thermal Pressure

Architecture	Thermal Pressure (α / β)	Crossover Condition	Trigger Point (q_{avg}/q_{wall})	Role of λ
Uniform CPU	Low (< 2)	$R_{ext} \propto \lambda^{1/4}$	Late: > 0.9	Safety Margin: High λ allows early engagement to keep fans quiet; low λ delays use until the wall.
High-Perf GPU	Mid (5 – 8)	$R_{ext} \approx \sqrt{\lambda} R_{int}$	Middle: 0.7 – 0.8	Operating Gear: λ defines the "Efficiency Floor" that prevents the cubic COP collapse.
AI Accelerator	High (10 – 15)	$R_{ext} \ll R_{int}$	Early: 0.4 – 0.6	Primary Engine: High α/β forces engagement even if λ is small.
3D Stacked IC	Extreme (> 20)	$R_{ext} \rightarrow 0$	Immediate: < 0.2	Survival Factor: λ is the only metric of success; bulk cooling is irrelevant.

What's missing, still?
Leakage power!

Chip power distribution	$Q = Q_{dyn} + Q_{leak}$
Hot spot dynamic power fraction	$\alpha_{dyn} = \frac{Q_{dyn,H}}{Q_{dyn}}$
Hot spot leakage power fraction	$\alpha_{leak} = \frac{Q_{leak,H}}{Q_{leak}}$

The key leakage-savings regime is not just high $\gamma_H = \alpha/\beta$. It is:

$$\beta \ll 1, \quad \alpha_{dyn} \ll 1, \quad \gamma_H \gg 1, \quad \alpha_{leak}/\alpha_{dyn} > 1.$$

That means the hotspot is a small dynamic-power region but becomes disproportionately expensive when hot because leakage concentrates there. Then cooling below the limit can reduce Q_{leak} , whereas in the fixed- ΔT wall comparison there is no separate leakage saving.

Rank	Chip type	β : hotspot area / die area	α_{dyn} : hotspot dynamic power / total power	$\alpha_{leak}/\alpha_{dyn}$: hotspot leakage power / dynamic power	Typical junction temperature T_j where leakage matters	Leakage-saving relevance for MR
1	NAND / low-power memory	0.2–0.6	0.2–0.6	0.2–0.6	40–70°C	Low; leakage is usually not the limiting thermal term
2	DRAM / commodity memory	0.1–0.4	0.2–0.5	0.5–1.5	50–85°C	Moderate; retention/leakage improves with cooling, but hotspots rarely dominate COP
3	HBM stack	0.05–0.25	0.2–0.6	0.7–2	70–95°C	Moderate-to-high; stacked thermal resistance makes leakage feedback more relevant
4	Mobile SoC	0.02–0.15	0.2–0.5	1–3	60–95°C	High; leakage can be a major part of thermally limited burst operation
5	Server CPU	0.02–0.10	0.2–0.5	1–4	75–105°C	High; local cooling can reduce leakage in boost-limited regions
6	GPU / AI accelerator	0.05–0.20	0.3–0.7	0.8–3	70–95°C	High, but (α) may be large, so MR energetic burden can also be large
7	Chiplet / 2.5D HPC package	0.01–0.10	0.2–0.6	1–4	75–105°C	High; leakage and package nonuniformity can both favor local cooling
8	3D logic-on-logic / stacked compute	0.005–0.05	0.2–0.6	2–6	80–110°C	Very high; leakage feedback can materially amplify MR value
9	RF / GaN / SiC power IC	0.001–0.03	0.2–0.8	Not same CMOS metric; use loss/leakage ratio	100–200°C+	Cooling helps efficiency/reliability, but model should use temperature-dependent conduction/switching/leakage loss rather than CMOS leakage ratio

Dynamic/Leakage Power Model

Dynamic Power	$Q_{dyn} = \alpha_{sw} C_{eff} V^2 f$
α_{sw}	Switching activity factor
C_{eff}	Effective switched capacitance
V	Supply dynamic voltage
f	Clock frequency

Frequency Voltage Model	$f(V) = K_f \frac{(V - V_{th})^\nu}{V}$
K_f	Transistor/geometric design constant
ν	Saturation velocity power exponent ~ 1–2
V_{th}	Critical path effective threshold voltage

Dynamic Voltage Model	$Q_{dyn} = K_D V(V - V_{th})^\nu$
K_D	$K_D = \alpha_{sw} C_{eff} K_f$

Leakage Current	$I_{sub}(V, T) = I_0 W_{eff} \exp\left(\frac{\eta_D V - V_{th}}{n V_T}\right) \left[1 - \exp\left(-\frac{V}{V_T}\right)\right]$
I_0	Process/device prefactor
W_{eff}	Effective total off-device width
n	Subthreshold slope factor
V_T	Thermal voltage $V_{th} = \frac{kT}{q}$
η_D	DIBL coefficient
V	Drain-source voltage ~ supply dynamic voltage

Leakage Power	$Q_{leak} = V I_{leak}(V, T)$ $\approx Q_{leak,0}(V) \exp[\kappa(T - T_{ref})]$
I_{leak}	$I_{leak}(V, T) = I_{sub}(V, T) + I_{gate}(V) + I_{junc}(V, T)$
$Q_{leak,0}(V)$	$V I_0 \exp\left(\frac{\eta_D V - V_{th}}{n V_{Tref}}\right)$

Workload-Agnostic Chip-Level Power Distribution Model

It is unrealistic to assume that all increase in chip power comes from frequency scaling (higher voltage), but rather it is a mixture of several performance/TDP scaling knobs

Scaling Knob	Design Impact	Effect on Q_{dyn}	Effect on Q_{leak}
$C_{eff} \uparrow$	More active area/units	Linear	Linear
α_{sw}	Higher utilization (lower dark silicon)	Linear	Weak
$f \uparrow$	Higher frequency	Linear	Requires $V \uparrow$
$V \uparrow$	Higher voltage	Quadratic	Exponential
$V_{th} \downarrow$	Lower threshold voltage / faster cells / better energy-delay product	Nonlinear (speed)	Exponential

Scaling scenario	Assumption	Use Case
Area/utilization	V, f, V_{th} fixed	GPUs/AI accelerators under parallel gains
DVFS boost	$f \uparrow, V \uparrow$	CPU/GPU clock boost
Energy-delay (leakage-limited)	$V_{th} \downarrow \Rightarrow$ higher T but leakage fraction growth	Advanced nodes / 3D logic

Workload-Agnostic Chip-Level Power Distribution Model

Normalizing DVDF to a reference voltage

and frequency operating point $V_0, f_0 \Rightarrow \frac{f(V)}{f_0} = \frac{V_0}{V} \left(\frac{V - V_{th}}{V_0 - V_{th}} \right)^v$

Dynamic power	$Q_{dyn}(\lambda) = \lambda_c(\lambda) C_0 V(\lambda)^2 f(\lambda)$	Useful switching power
Leakage power	$Q_{leak}(\lambda) = \lambda_A V(\lambda) I_{leak}(V, T)$	Parasitic power

Scaling Parameters	Equation	Meaning
Performance knob	λ	Desired workload/performance scaling
Capacitance & Leakage-Area Scaling	$\lambda_c(\lambda) = \lambda^a, \quad \lambda_A(\lambda) = \lambda^b$	More active capacitance / parallelism; Powered leakage area / device width scaling
Frequency scaling relation	$f(\lambda) = f_0 \lambda^{1-a}$	Remaining scaling comes from clock
DVFS scaling	$\lambda^{1-a} = \frac{V_0}{V} \left(\frac{V - V_{th}}{V_0 - V_{th}} \right)^v$	Solve for $V(\lambda)$
Thermal voltage	$V_T = \frac{kT}{q}$	Thermal voltage ~26 mV at 300 K
Threshold voltage	V_{th}	Device threshold voltage
Leakage current	$I_{leak} = I_0 \exp\left(\frac{\eta_D V - V_{th}}{n V_T}\right)$	Compact subthreshold/DIBL leakage
Dynamic power	$Q_{dyn} = C_0 V_0^2 f_0 \lambda \left[\frac{V(\lambda)}{V_0} \right]^2$	
Leakage power	$Q_{leak} = \lambda^b V(\lambda) I_0 \exp\left(\frac{\eta_D V - V_{th}}{n V_T}\right)$	
Hot-spot power fractions	$\alpha_{leak} = \frac{Q_{leak,H}}{Q_{leak}}, \quad \alpha_{dyn} = \frac{Q_{dyn,H}}{Q_{dyn}}$	Leakage localization
Total hot-spot fraction	$\alpha_{tot} = \frac{Q_{leak,H} + Q_{dyn,H}}{Q_{leak} + Q_{dyn}}$	Effective thermal hot-spot power fraction
Thermal pressure	$\gamma_H = \frac{\alpha_{tot}}{\beta}$	Thermal pressure by the hot spot

Workload-Agnostic Chip-Level **Power Distribution Model**

Comparison	Scaling Knob	Leakage treatment
Max TDP / thermal wall	Hold hot spot temperature constant ($\Delta T_H = \Delta T$)	No leakage saving; use calibrated α_{tot}
Same workload, cooler hybrid	Hold Q_{dyn} fixed but decrease ΔT_H	Leakage decreases with lower T_H
Future architecture scaling	Vary λ	Use DVFS/leakage model to update chip power

Scaling Architecture Parameters
$Q_{dyn} = C_0 V_0^2 f_0 \lambda \left[\frac{V(\lambda)}{V_0} \right]^2$
$Q_{leak} = \lambda^b V(\lambda) I_0 \exp \left(\frac{\eta_D V - V_{th}}{n V_T} \right)$
$\lambda^{1-a} = \frac{V_0}{V} \left(\frac{V - V_{th}}{V_0 - V_{th}} \right)^v$

Scaling scenario	Assumption
More cores / larger die	$\lambda_C \uparrow, \lambda_A \uparrow$; V, f fixed
Frequency boost	$f \uparrow$; maybe $V \uparrow$
Voltage Boost	$V \uparrow$; strong leakage rise
Lower threshold voltage	$V_{th} \downarrow$; strong leakage rise

Future architecture	a	b	Physical interpretation
Same chip; more utilization	1	0	More switching in existing logic; leakage area unchanged
Same chip, more blocks active	1	0.3–0.7	Dynamic power rises from activating more units; some extra leakage area turns on
Bigger accelerator / more parallelism	1	1	More compute units, more switched capacitance, and more leakage area
Ideal power-gated scaling	1	$\ll 1$	Added parallelism only leaks when active; standby leakage mostly suppressed
CPU boost / same core faster	0	0	Same capacitance and leakage area; higher (f) requires higher (V)
Aggressive single-thread boost	0–0.3	0	Mostly DVFS; voltage and leakage penalty are strong
Mixed GPU / AI scaling	0.6–0.9	0.5–1	Mostly parallelism, some clock/voltage increase, leakage area grows too
Chiplet scaling	0.7–1	0.6–1	More chiplets/tiles add capacitance and leakage area; modest DVFS
3D stacked logic	0.5–1	0.5–1	More active silicon plus higher thermal leakage feedback
Low-threshold binning	N/A	N/A	a, b alone are insufficient; also reduce V_{th} causing leakage jump

Workflow:

Power-Chip Calibrated Bottom-Up Model

Stage	Required inputs / equations	Output
1. Reference state	$V_0, f_0, Q_{\text{dyn},0}, Q_{\text{leak},0}$	Baseline chip state at $\lambda = 1$
2. Technology inputs	$V_{\text{th}}, \nu, I_0, n, \eta_D, V_t$	DVFS + leakage model
3. Hotspot inputs	$\beta, \alpha_{\text{dyn},0}, \alpha_{\text{leak},0}$	Reference spatial power localization
4. Scaling path	λ, a, b	Future workload / architecture path
	$f(\lambda) = f_0 \lambda^{1-a}$	Frequency scaling
	$\lambda^{1-a} = \frac{V_0}{V} \left(\frac{V - V_{\text{th}}}{V_0 - V_{\text{th}}} \right)^\nu$	Solve for $V(\lambda)$
5. Dynamic power	$Q_{\text{dyn}} = Q_{\text{dyn},0} \lambda \left(\frac{V}{V_0} \right)^2$	Scaled dynamic power
6. Leakage power	$Q_{\text{leak}} = Q_{\text{leak},0} \lambda^b \frac{V}{V_0} \exp \left[\frac{\eta_D (V - V_0)}{n V_t} \right]$	Scaled leakage power
7. Total power	$Q = Q_{\text{dyn}} + Q_{\text{leak}}$	Total chip heat
8. Effective hotspot fraction	$f_{\text{leak}} = \frac{Q_{\text{leak}}}{Q}$	Leakage fraction
	$\alpha_{\text{tot}} = (1 - f_{\text{leak}}) \alpha_{\text{dyn},0} + f_{\text{leak}} \alpha_{\text{leak},0}$	Total hotspot power fraction
	$\gamma_H = \frac{\alpha_{\text{tot}}}{\beta}$	Hotspot severity
10. Cooling model	Use $Q, \alpha_{\text{tot}}, \gamma_H, \beta, COP_\mu$	Solve u^*, s^* , compute COP
	$COP_{\text{hyb}} = \frac{Q}{P_{\text{pump}}(u^*) + \alpha_{\text{tot}} s^* Q / COP_\mu}$	Hybrid COP

Architecture Reference Points

Architecture	V_0	f_0	$Q_{\text{leak},0}/Q_{\text{dyn},0}$	β	$\alpha_{\text{dyn},0}$	$\alpha_{\text{leak},0}$	(a, b) default
Mobile SoC	0.7–0.9 V	1–3 GHz	0.1–0.4	0.02–0.10	0.15–0.40	0.2–0.6	0.5–0.8, 0.3–0.7
General SoC / edge AI	0.7–0.9 V	0.5–2.5 GHz	0.1–0.3	0.03–0.15	0.15–0.50	0.2–0.6	0.7–1.0, 0.5–1.0
Server CPU	0.8–1.1 V	2–5 GHz	0.2–0.7	0.01–0.08	0.20–0.50	0.3–0.8	0.2–0.6, 0.2–0.6
GPU / AI accelerator	0.7–1.0 V	1–2.5 GHz	0.1–0.4	0.05–0.20	0.25–0.70	0.3–0.8	0.7–1.0, 0.7–1.0
SRAM / cache-heavy block	0.6–0.9 V	tied to logic	0.3–1.0+	0.02–0.10	0.10–0.40	0.3–0.9	0.5–1.0, 0.8–1.0
Chiplet / 2.5D HPC	0.7–1.0 V	1–4 GHz	0.15–0.6	0.01–0.10	0.20–0.60	0.3–0.8	0.6–1.0, 0.5–1.0
Advanced 3D logic	0.6–0.9 V	1–3 GHz	0.3–1.0+	0.005–0.05	0.20–0.60	0.4–0.95	0.5–1.0, 0.5–1.0

Useful defaults for unknowns:

$$\nu \sim 1.2\text{--}2, \quad n \sim 1.2\text{--}1.8, \quad \eta_D \sim 0.05\text{--}0.15, \quad V_t \simeq 26 \text{ mV at } 300 \text{ K}.$$

Architecture	V_0	f_0	β	$\alpha_{sw,H}/\alpha_{sw,B}$	$V_{th,H} - V_{th,B}$	Leakage character	Typical scaling scenario $((a, b))$	Comments
Mobile SoC	0.7–0.9 V	1–3 GHz	0.02–0.10	3–10	Slightly lower V_{th} in performance cores	Moderate leakage	(0.5–0.8, 0.3–0.7)	Mixed DVFS + parallelism; thermal throttling important
Laptop / desktop CPU	0.8–1.2 V	3–6 GHz	0.01–0.05	5–20	Often lower V_{th} in critical paths	High leakage at boost	(0–0.4, 0–0.3)	Strong DVFS boost behavior
Server CPU	0.8–1.1 V	2–5 GHz	0.01–0.08	4–15	Aggressive low- V_{th} libraries	High leakage	(0.2–0.6, 0.2–0.5)	Balance of DVFS + multicore scaling
GPU / AI accelerator	0.7–1.0 V	1–2.5 GHz	0.05–0.20	2–8	Less aggressive V_{th} binning	Moderate-to-high leakage	(0.7–1.0, 0.7–1.0)	Mostly parallelism scaling
Inference ASIC / TPU-like	0.6–0.9 V	0.5–2 GHz	0.05–0.15	2–6	Moderately optimized V_{th}	Moderate leakage	(0.8–1.0, 0.8–1.0)	Area scaling dominates
HBM / stacked memory	0.6–0.9 V	1–3 Gbps/pin equiv.	0.05–0.25	1.5–5	Usually conservative V_{th}	Moderate leakage	(0.8–1.0, 0.8–1.0)	Thermal coupling dominates
SRAM / cache arrays	0.6–0.9 V	tied to logic clock	0.02–0.10	2–5	SRAM cells use higher V_{th}	Leakage sensitive	(0.5–1.0, 0.8–1.0)	Leakage becomes large at advanced nodes
Advanced 3D logic	0.6–0.9 V	1–3 GHz	0.005–0.05	5–30	Often aggressive low- V_{th}	Very high leakage sensitivity	(0.5–1.0, 0.5–1.0)	Strong thermal feedback and hotspot amplification
Chiplet / 2.5D HPC	0.7–1.0 V	1–4 GHz	0.01–0.10	3–10	Mixed V_{th} libraries	Moderate-to-high leakage	(0.6–1.0, 0.5–1.0)	Parallelism + packaging thermal limits
RF / mixed-signal	1–5 V	MHz–100s GHz	0.001–0.05	highly localized	Technology-specific	Not CMOS-leakage dominated	scenario-dependent	Often dominated by self-heating/reliability rather than digital leakage
GaN / SiC power IC	5–1000 V	kHz–MHz	0.001–0.03	highly localized	Not CMOS V_{th} scaling	Conduction/switching losses dominate	scenario-dependent	Use electrothermal loss models instead of CMOS leakage

ROADMAP & CORE ARGUMENTS

Why Now?

An overview of why photonic cooling is no longer a luxury, but a physical necessity for the next generation of microelectronics.

Hot-Spot Density Rules

Modern microprocessors dissipate +10 **W/mm²** in micron-scale, sub-microsecond bursts.

The active-layer flux —not total chip power —is the binding constraint for performance.

The Conduction Wall

Existing cooling (Air → Liquid → Two-phase) is bounded by the **conduction bottleneck** through the lid, TIM, and cold plate.

ASF microrefrigeration acts as a thermal shunt, bypassing the passive resistance stack.

PCP Operating Regimes

Hybrid Clipping: Relieves fans.

R = 0: Zero die resistance.

R < 0: Over-cooling.

Energy recovery efficiency $\eta > 70\%$.

Two Paths to Microrefrigeration: TEC vs. Laser Photonic Cooling

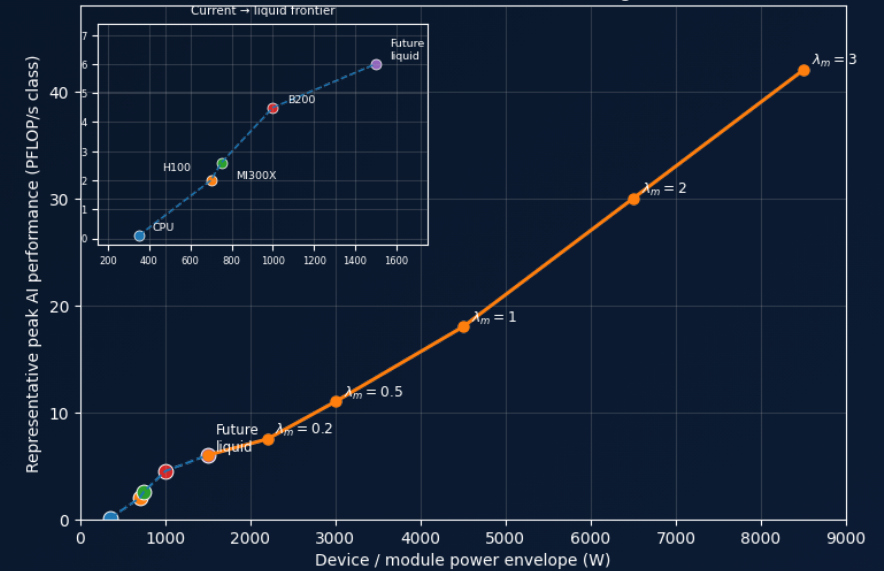
⚡ THERMOELECTRIC COOLER (TEC)

- ✓ **Mechanism:** Peltier effect —electrical current drives electron-phonon heat pumping across a p-n junction.
- ✓ **Maturity:** COTS modules available (Bismuth Telluride), wafer-bonded thin-film variants emerging.
- ✓ **COP:** Typically 0.2–0.8 at $\Delta T = 20\text{--}60^\circ\text{C}$. Limited by $ZT \approx 1$ (Bi_2Te_3).
- ✗ **Limitation:** Joule heating adds to the load —COP degrades rapidly with temperature lift. Phonon backflow limits minimum achievable T.
- ✗ **Scale:** Minimum footprint $\sim 0.5\text{mm}^2$. Difficult to scale below $100\text{ }\mu\text{m}$ pitch without contact resistance dominance.

🔬 LASER / PHOTONIC COOLING (LUMINESCENCE UPCONVERSION)

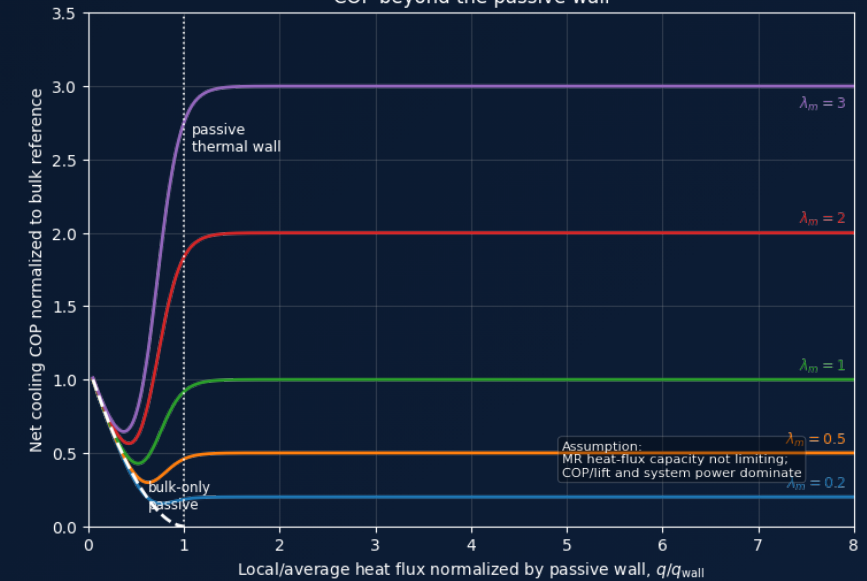
- ✓ **Mechanism:** Anti-Stokes luminescence —each absorbed photon is re-emitted at higher energy, carrying away phonon heat.
- ✓ **COP:** Can exceed Carnot efficiency for local junction —entropy is carried away by photons, not electrons.
- ✓ **Contact-Free:** No electrical contact to cooled region —immune to contact resistance scaling issues.
- ✓ **Platform:** Rare-earth doped crystals, molecular dispersions, III-V semiconductors (GaAs, InP). Photonic engineering optimizes emission directionality.
- ✗ **Challenge:** Requires background absorption suppression and precision laser pump delivery. Material platform engineering is the current bottleneck.

Thermal wall → MR-enabled scaling



Microrefrigeration projection with effectively unlimited local heat-flux capacity

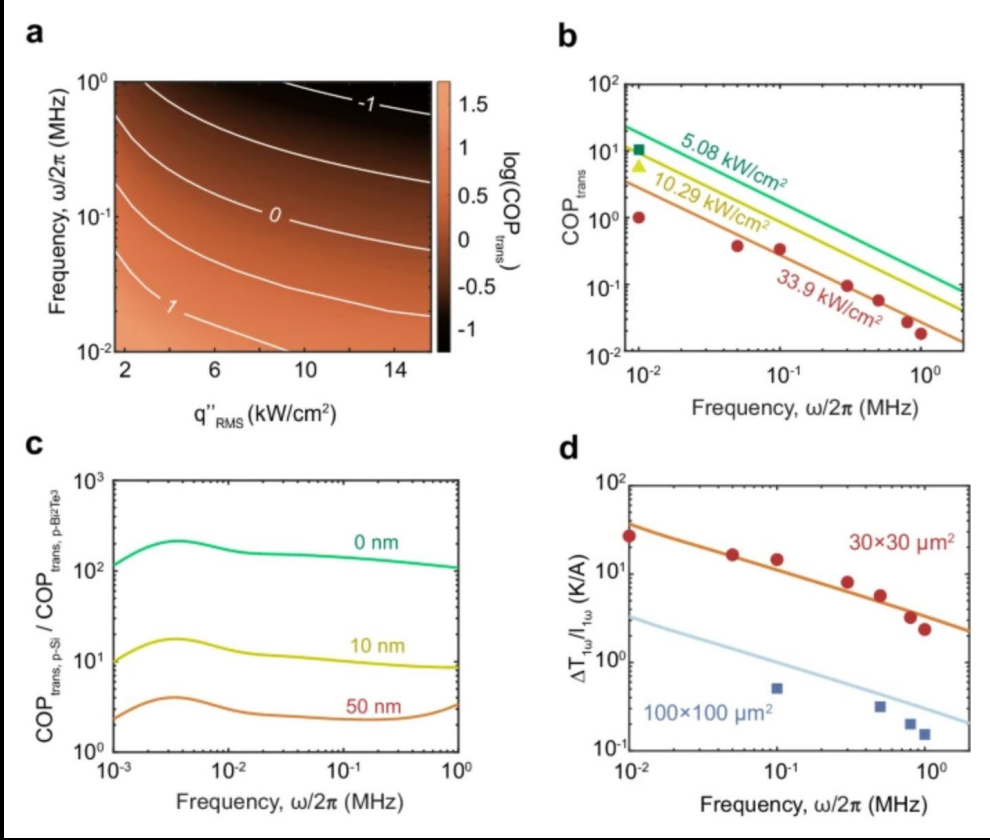
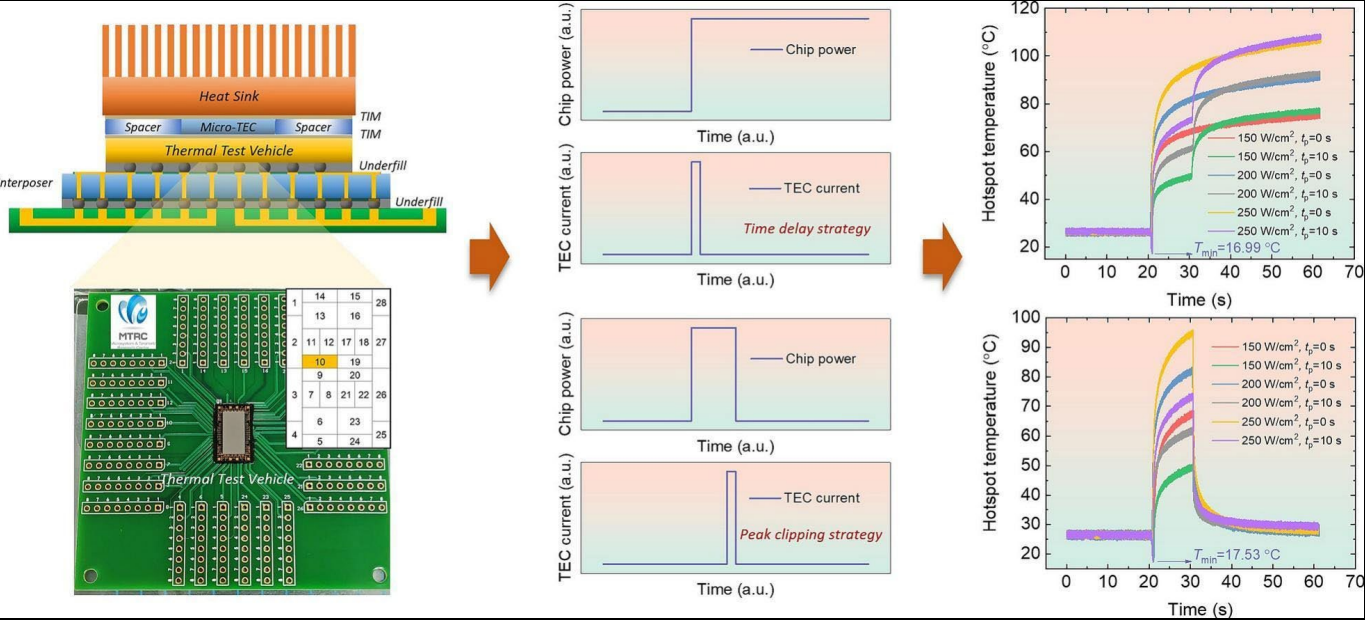
COP beyond the passive wall



Upshot: Even “bad” microrefrigerator COP is acceptable when it attacks only the **temperature excess (“clipping”)**, not the full hotspot heat load.

What’s so special about Laser Cooling Microrefrigeration?

- **Thin-Film On-Chip Thermo-Electric Cooling (μ TEC)**
 - Demonstrated meaningful **peak clipping** and performance gains under dynamical spatial ($\lesssim 100\mu\text{m}$) and temporal targeting ($\lesssim \text{ms}$)
 - Suffers from significant parasitic resistance: **COP (practical) ~ 0.3**
 - Bulk cooler needs to reject MR waste heat (further waste)
 - Maximum cooling power density $\frac{S_H}{A_H} \sim 1 - 10 \text{ W/mm}^2$ and $\Delta T_{\text{trim}} \lesssim 10 \text{ K}$



Upshot: Even “bad” microrefrigerator COP is acceptable when it attacks only the **temperature excess (“clipping”)**, not the full hotspot heat load.

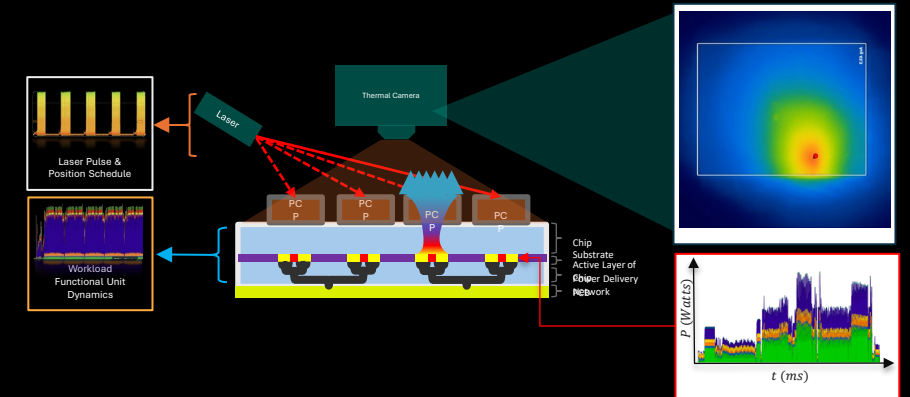
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- **Solid-State Thin-Film On-Chip Laser Cooling (LC)**
 - Unprecedented (orders of magnitude greater) dynamical spatial ($\leq \mu\text{m}$) and temporal ($< \mu\text{s}$) targeting resolution
 - No electrical components $\Rightarrow$ no parasitic resistance
 - Bulk cooler does not need to reject MR waste: instead, extracted heat “recycled” via laser power converters (LPC)
 - Intrinsic COP $\sim 0.1\text{-}0.5$ can be further increased through LPC thermodynamic cycle (optimization):
 - COP (theory): technically infinite and even negative (net power generation = energy harvesting)
 - **COP (practical): $\sim > 1$** , limited by laser wall-plug efficiency under cooling integration $\sim 65\text{-}70\%$, LPC efficiency $\sim 80\text{-}90\%$
 - Maximum cooling power density $\frac{S_H}{A_H} \sim 10 - 100 \text{ kW/mm}^2$ and $\Delta T_{trim} \lesssim 100\text{K}$ set by phonon relaxation rate & material purity

Keeps T of chip constant
for all power densities. This impacts:

Spatially: microns to mm
scales – matched hotspot
size

Temporally: nanoseconds
to match onset of heating
for workload



Performance

- Energy-delay-product (EDP)
- Targeted Frequency Scaling
- Static power regularization
- Memory / cache shielding
- Eliminate hotspot thermal throttling
- Reduce time-until-hotspot
- Higher transistor e^- mobility

Energy

Reduction / Recovery

- 1) Leakage Power (chip-injection)
- 2) Convective cooling (fan/pump/chillers)

Power Generation

- 1) Partial Waste-heat power recovery

30% efficiency

60% efficiency

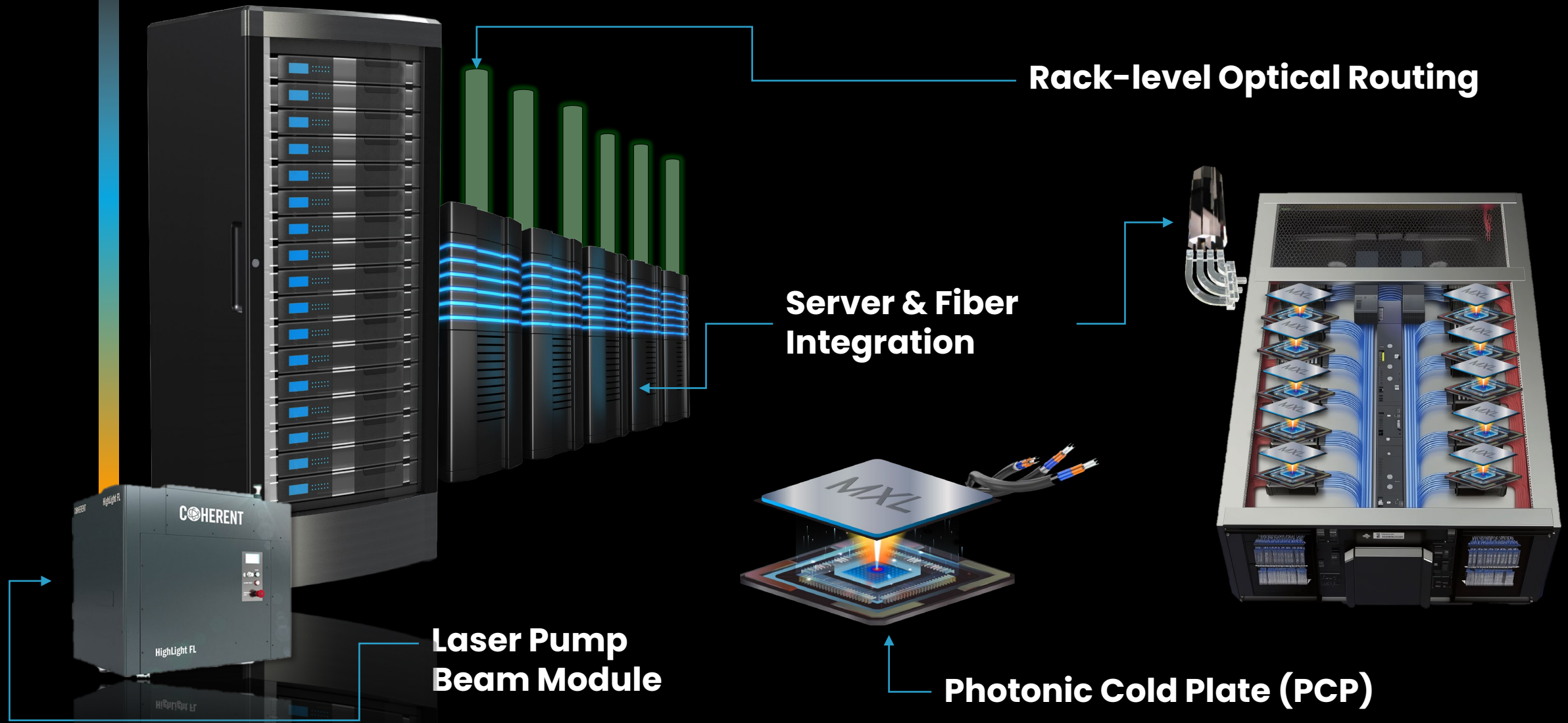
100% - **Breakeven**

160% efficiency

- 2) Net Power generation

PHOTONIC COOLING SOLUTION

End-to-End Photonic Cooling for HPC | **Solution Components**



Core Innovation >> MXL's Revolutionary Photonic Cold Plate

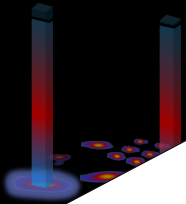
Photonic Network

Waveguide – Optical Bus &
Pixel Array

Photonic Network

Hotspots

Processor



50 mm

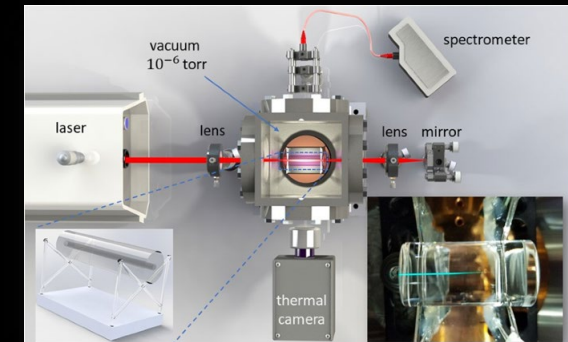
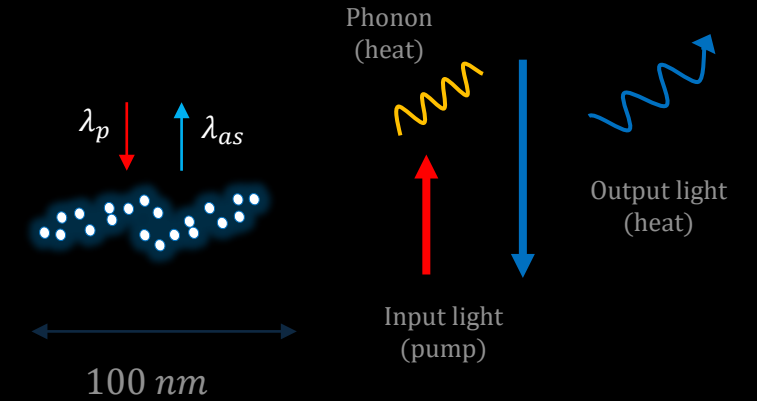
Thin-Film Cooling Pixel

Inverted Thin-Film Cooling Device



Reverse Laser Cooling

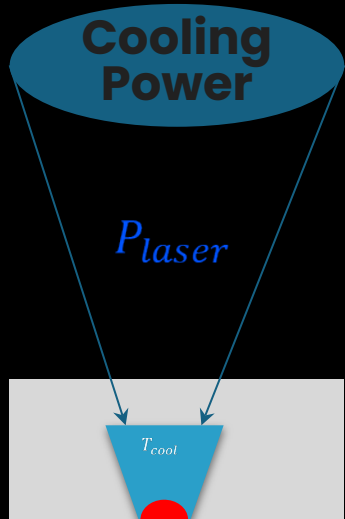
Anti-Stokes Fluorescence



Sarma, Rodriguez et al 2024

Albrecht et al UNM, 2020/2024

LASER COOLING? COP & POWER DENSITY



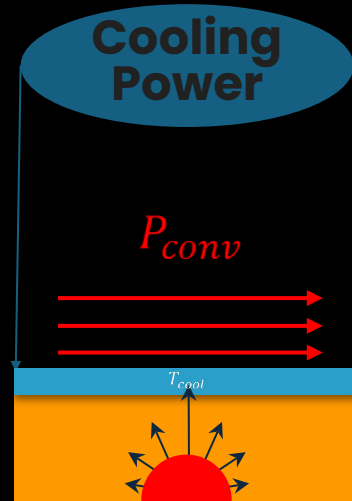
$$T_{cool} \sim P_{laser}$$

Laser Cooling

Light penetrates through substrate

T_{cool} is set by laser power

Light localizes to any length scale (microns to mm)



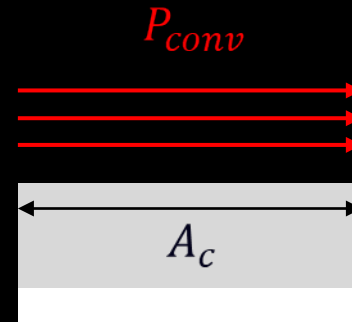
$$T_{cool} \sim T_{liquid}$$

Liquid Cooling

Fluid lifts from the surface, limited by conduction

T_{cool} set by **fluid temperature**

Cooling power delocalized

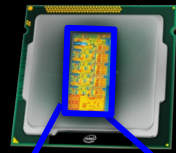


Large-Area Limit

(Low "Average" Power Density)

Convection offers area scaling with high-efficiency (COPs ≥ 3)

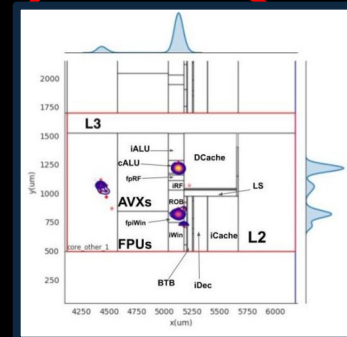
$$A \sim A_c$$



Laser Cooling offers Highly Localized Cooling

(Extreme "Average" Power Density)

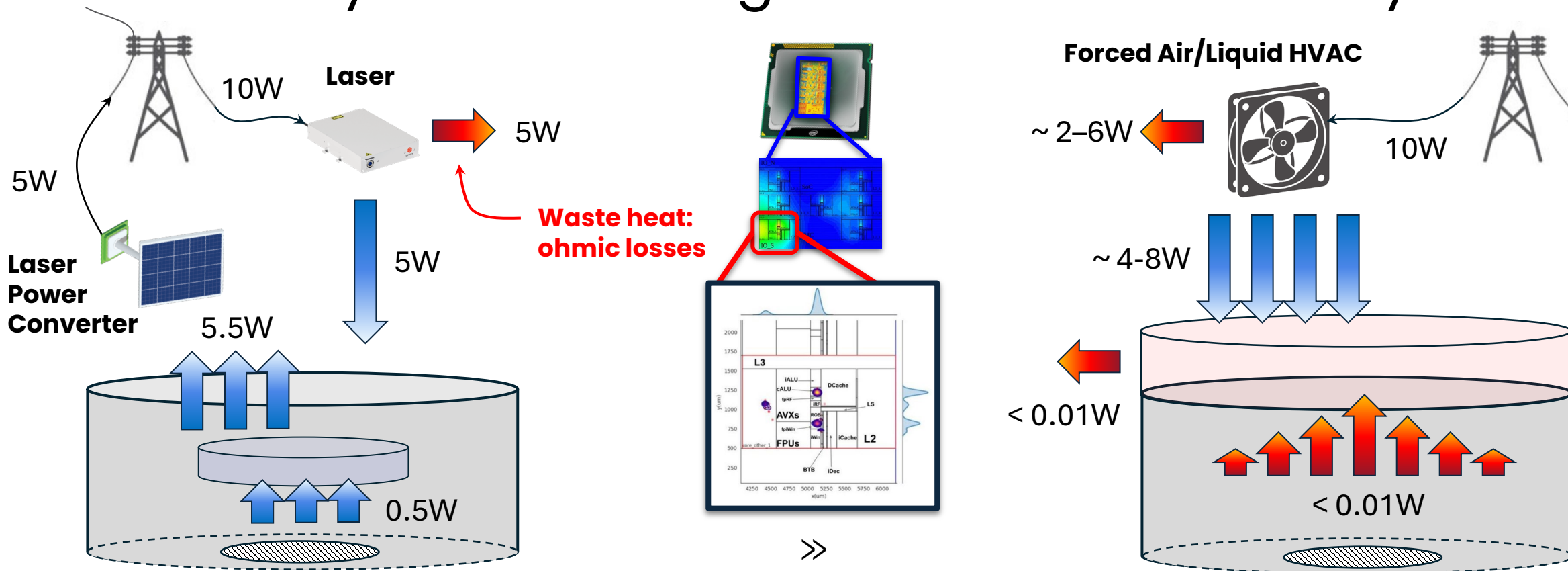
Laser cooling offers localized cooling with high-efficiency



Laser = Surgical Tool

Convection = Blunt Instrument

Why Laser Cooling? COP & Power Density



Laser Cooling offers Highly Localized Cooling

(Extremely High "Average" Power Density)
high-efficiency localized cooling

Light penetrates through wafer substrate $\Leftarrow$
 T_{cool} is set by laser power $\Leftarrow$
 Spatial (μm) and temporal (μs) localization $\Leftarrow$

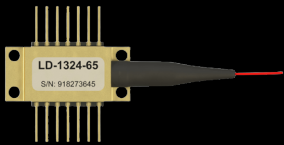
Convective Cooling offers Large-Area, Delocalized Cooling

(Low "Average" Power Density)
high efficiency with area scaling (COPs ≥ 3)

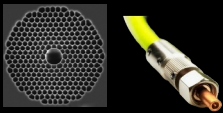
$\Rightarrow$ Lifts heat from surface
 $\Rightarrow$ Limited by conduction "resistance" and T_{cool}
 $\Rightarrow T_{cool}$ is fluid temperature

Commercialization Roadmap >> Fabrication & Assembly

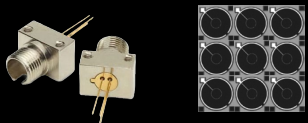
Commercially-Sourced



High-efficiency, high-power diode lasers



Hollow-core fiber (100W / fiber)



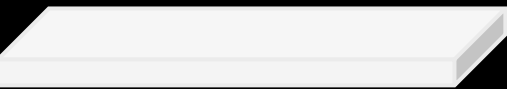
High efficiency Laser-Power Converters



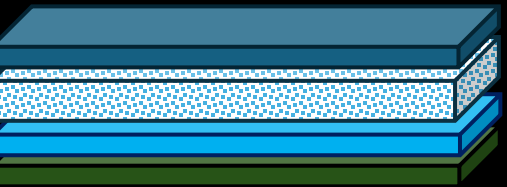
Pump & Idler Laser System



Optical coupling to fiber



Photonic Integrated Circuit & waveguide networks



Laser-power converter

Coupler Inverse-designed structure

Extractor layer for up-conversion

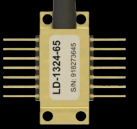
Type III-V Bragg Reflectors

Thermo-optic Thin Films

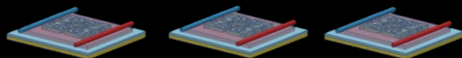
Photonic Network Fabrication



LPC, Fiber Integration



Pulsed Laser integration



Embed PCP tiles in thin-film cold plate mount

In-House Bonding & Assembly



High-efficiency materials Optimization

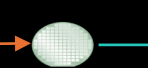


Coupler Wafer

Extractor Boule

DBR Wafer

Sensor Deposition



Wafer-bond and dice PCP tiles

Outsourced Wafer Fabrication

The Conduction Bottleneck: Why Microrefrigeration Lights the Way

“ In this architecture, we use ASF laser cooling to create a **'Thermal Shunt.'** By extracting phonon energy at the hot spot and converting it to photons that exit the package at the speed of light, we are not cooling the chip —we are **optically screening (even cancelling) the thermal resistance** of the die. This allows one to operate at a **'Null Resistance' point** where the hot spot is thermally transparent, preventing the bulk system from ever entering its inefficient, high -speed cubic regime.

THREE OPERATING REGIME PREVIEW

I THERMAL CLIPPING

$0 < S < S_{null}$

MR partially cancels downstream resistance. Bulk cooling fans slow down. Net efficiency improves.

II RESISTANCE NULLIFICATION

$S = S_{null}$

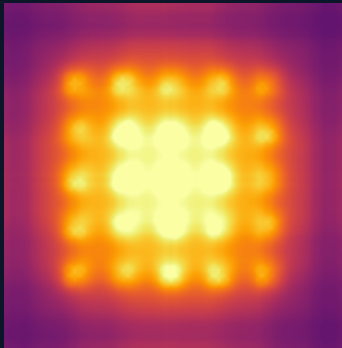
$R_{eff} = 0$. Junction temperature equals ambient. Die is thermally transparent. Optimal operating point.

III OVER-COOLING

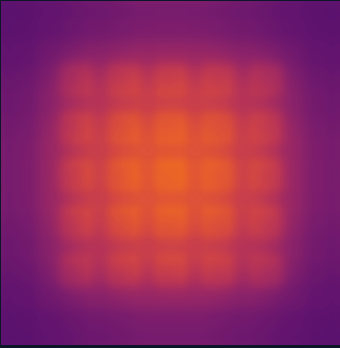
$S > S_{null}$

$R_{eff} < 0$. $T_j < T_a$. Die acts as a cold source. Net power generation becomes feasible with LPC.

💡 **Key Architecture Insight:** Only the microrefrigerator's strength S is optimized. The bulk system operates at a fixed point. MR acts as a **precision thermal compensator**, intercepting exactly the energy needed to keep $T_j \leq T_{lim}$.



Air-Cooling Only



Zero-Resistance Die



Over-Cooling

3. Regimes of *s*—How MR Tunes Apparent Thermal Resistance

FULL CANCELLATION POINT (DEFINITION)

$$s_0 = 1 + \frac{R_{die}}{R_{down}}$$

$s = s_0$

$R_{eff} = 0$ At and—zero-resistance die

$T_j = T_a$

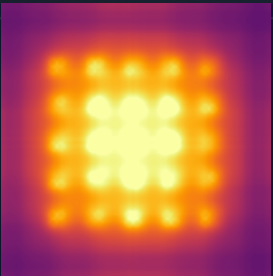
EFFECTIVE RESISTANCE (REMINDER)

$$R_{eff}(s) = R_{die} + (1 - s)R_{down}$$

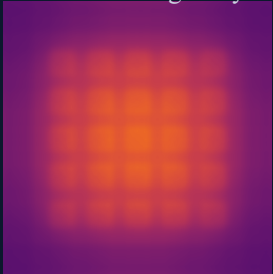
s linear in—each unit of MR strength cancels R_{down}

OPERATING REGIME MAP

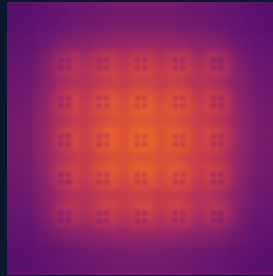
S RANGE	REFF	Tm VS Ta	PHYSICAL MEANING
$s = 0$	$R_{die} + R_{down}$	$T_m > T_a$	Passive thermal stack —no microrefrigerator active
$0 < s < 1$	$R_{die} + (1-s)R_{down}$	$T_m > T_a$	Thermal Clipping—partial downstream cancellation, fans slow down
$s = 1$	R_{die}	$T_m = T_a$	Hot-spot interception —full downstream cancellation, T_m pinned to ambient
$1 < s < s_0$	$0 < R_{eff} < R_{die}$	$T_m < T_a$	Resistance Nullification —partial die-resistance cancellation, sub-ambient MR node
$s = s_0$	0	$T_m = T_a - qR_{die}$	Zero-Resistance Die—full cancellation, $T_j = T_a$ exactly
$s > s_0$	$R_{eff} < 0$	$T_m < T_a - qR_{die}$	Over-Cooling —negative effective resistance, $T_j < T_a$, net power generation possible



Air-Cooling Only



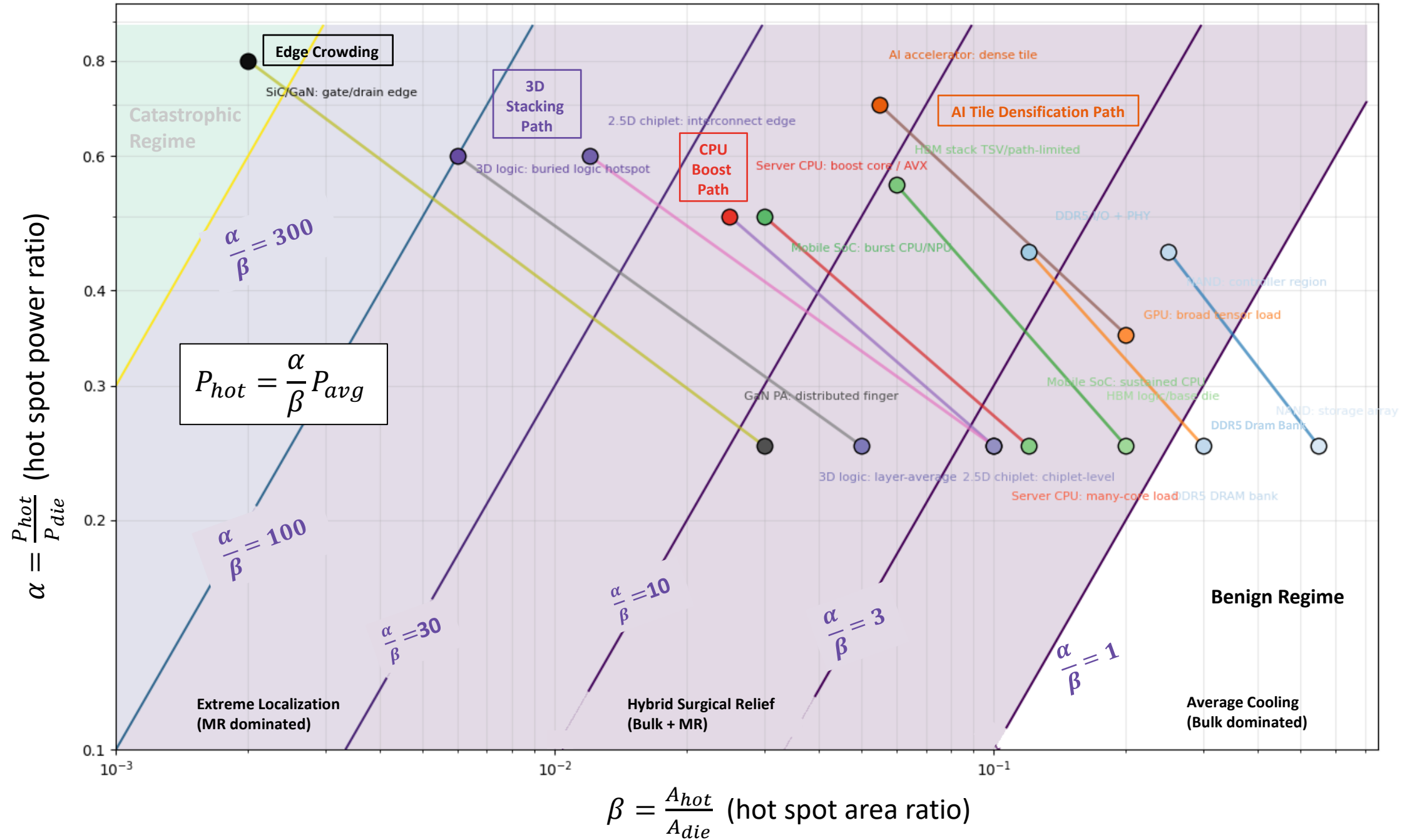
Zero-Resistance Die



Over-Cooling

At the optimum, whenever $s^* > 0$: T_j^* is pinned to T_{lm} (junction at thermal limit)

Thermal Pressure Phase Diagram



Back to Physics!

Cooling with Lasers: against popular conception



The Martians' heat-ray disperses the crowd.

From H.G. Wells *War of the Worlds*
Illustrated by Henrique Alvim Corrêa, 1906

Generally, lasers and electromagnetic radiation in general are thought of to heat things, and they do that quite well.

How would a laser cool something down?



While this laser cutter is carving a snowflake, it is doing so by superheating the material.

Click to edit Master title style

Example system:
Yb ions embedded in YLF glass

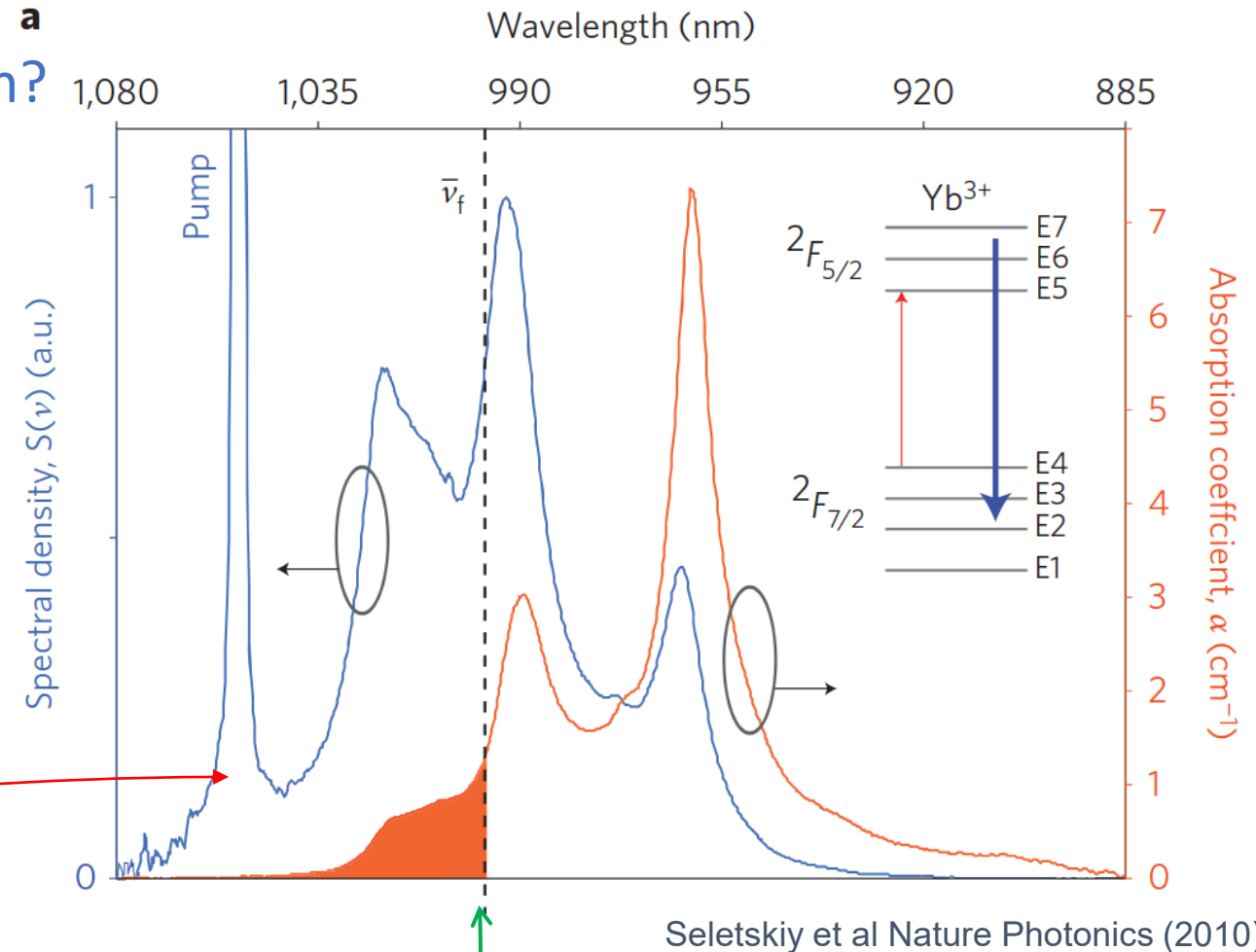
more energetic photons



How does shining a laser cool something down?

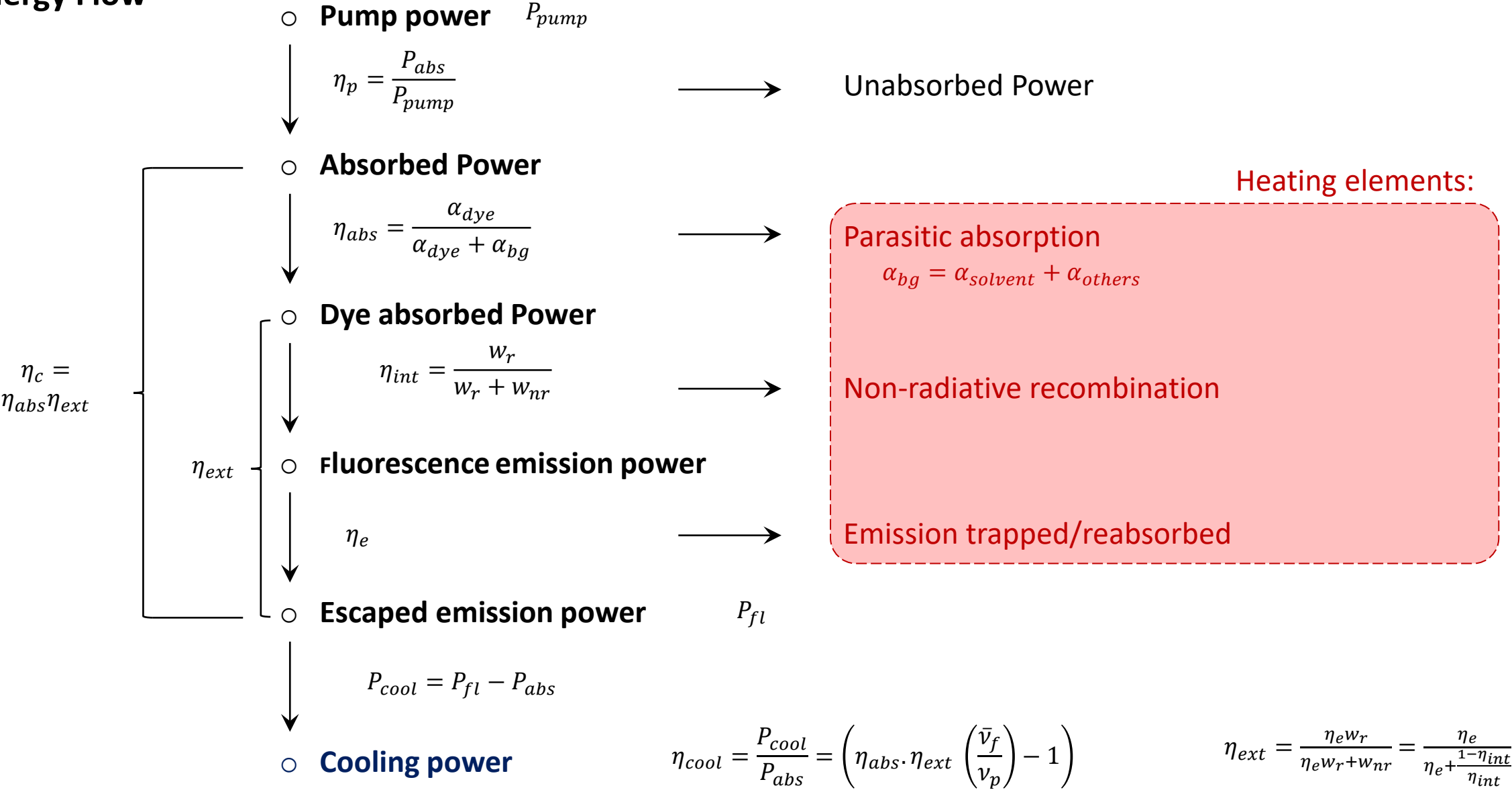
Cooling: more power is coming out (fluorescence) than going in (pump absorption)

Anti-Stokes fluorescence: fluorescence photons are on average more energetic than pump photons.



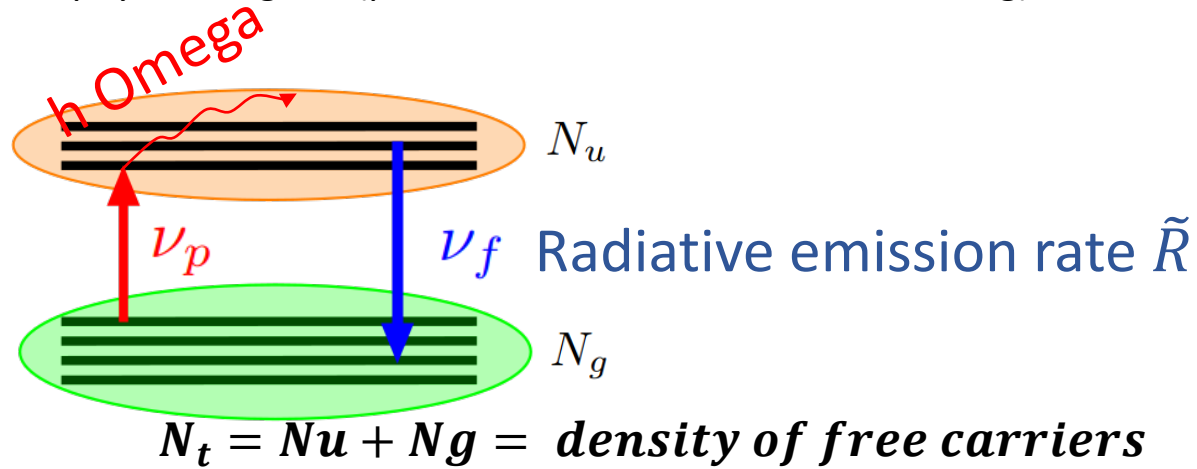
To observe this, must **pump** at lower energy than **mean fluorescence**

Energy Flow



What are the limitations in ASF cooling?

1) Basic physics diagram (phonon-assisted fluorescence cooling)

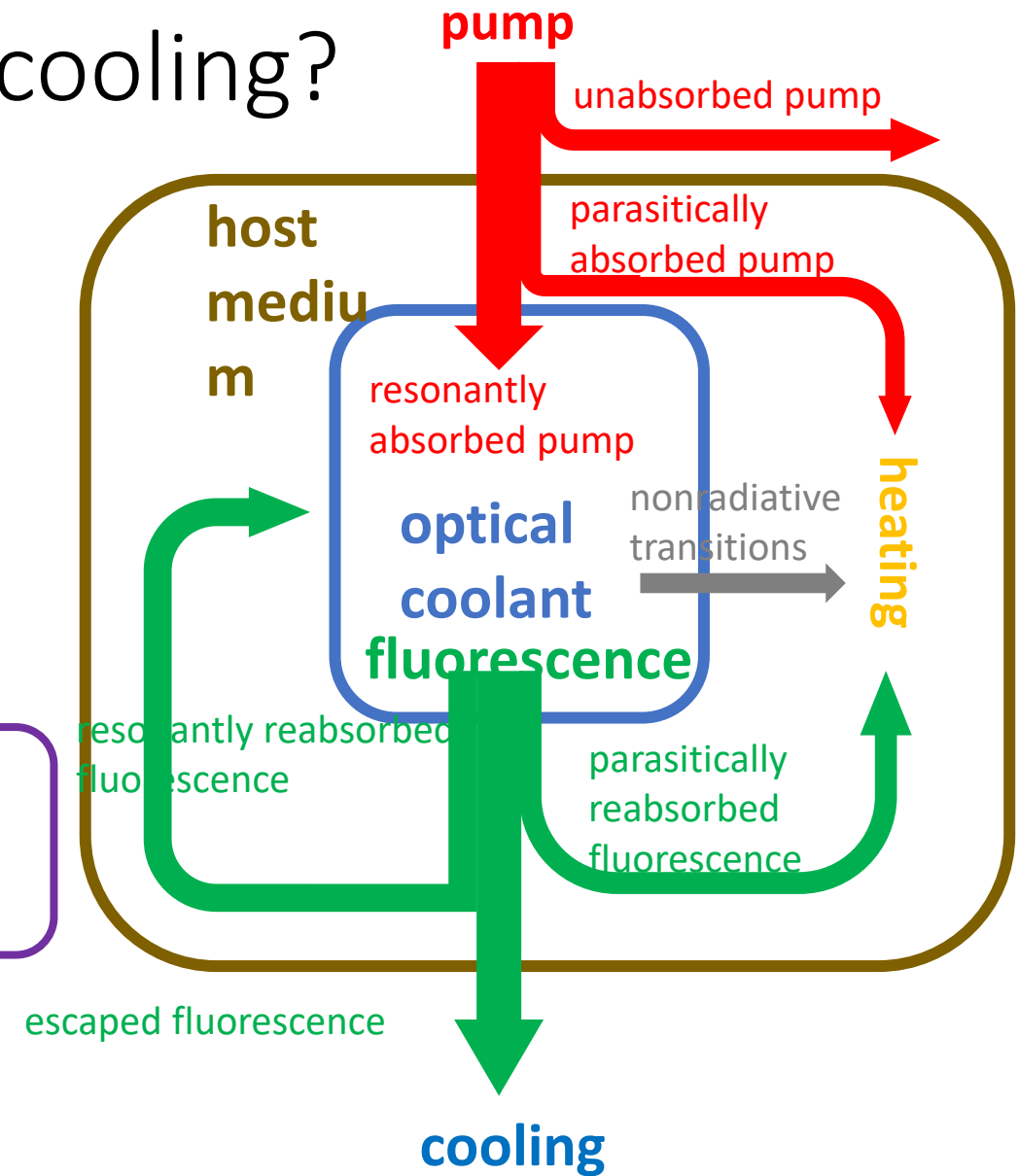


$$\eta_c = \eta_{abs} \eta_{EQE} \frac{\bar{\nu}_f}{\nu_p} - 1$$

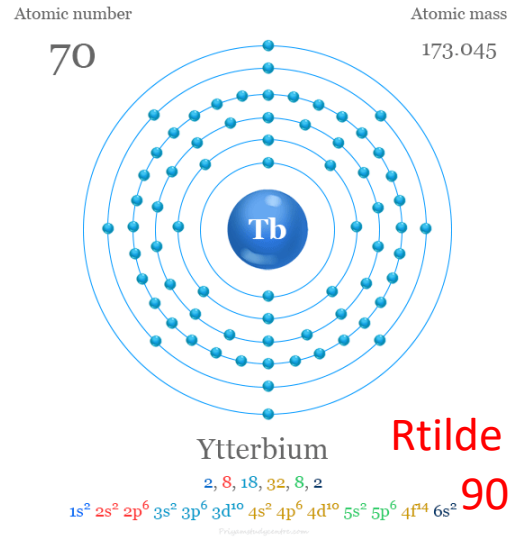
$$P_{cool,max} = \eta_c h \nu_p \tilde{R} N_t$$

$$\eta_{abs} = \frac{\alpha_{dye}}{\alpha_{dye} + \alpha_{bg}}$$

$$\eta_{ext} = \frac{\eta_e w_r}{\eta_e w_r + w_{nr}} = \frac{\eta_e}{\eta_e + \frac{1-\eta_{int}}{\eta_{int}}}$$



rare earth ions

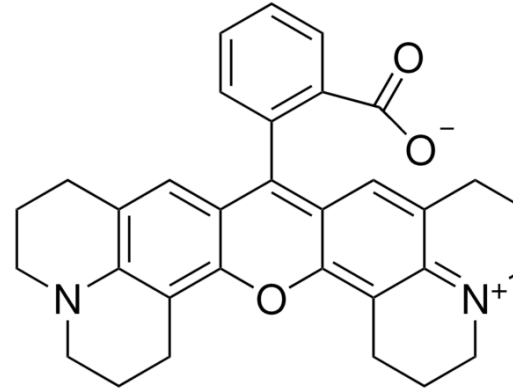


$\tau \sim \text{ms}$
900-1200nm

- dipole-forbidden 4f-4f transitions”
- High internal quantum efficiency
- Long radiative lifetime (ms scale) $\rightarrow$ low baseline max cooling power

“molecular” systems

fluorescent dyes

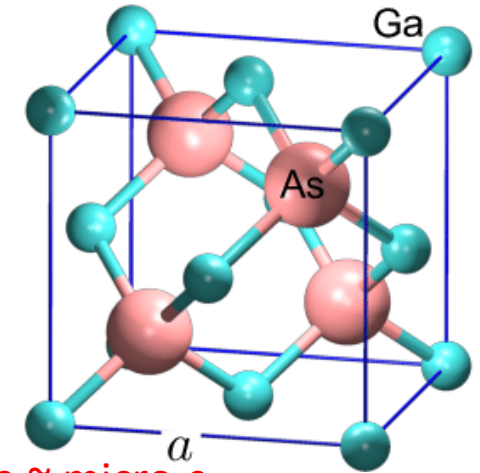


rhodamine 101

$\tau \sim \text{ns}$
500-850nm

- Molecular electronic state transitions
- High internal quantum efficiency for individual dye molecules, fast radiative transitions (ns scale)
- Internal quantum efficiency declines precipitously past certain concentration level

semiconductors

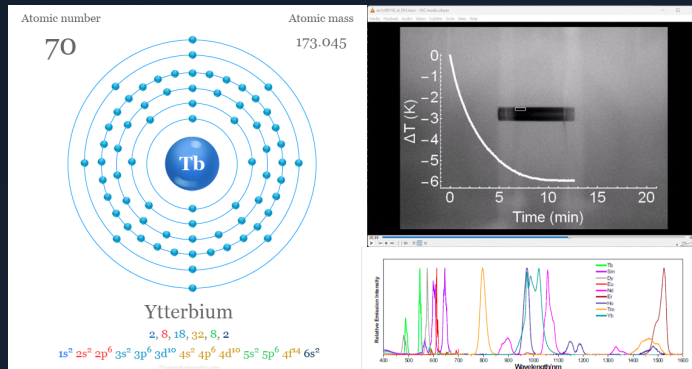


XXXnm?

$\tau \sim \text{micro-s}$

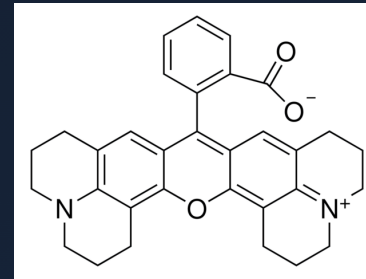
- Charge carrier excitation – relaxation between bands
- fast radiative transitions (ns scale), Fermi-Dirac statistics $\rightarrow$ potential improved cryogenic performance, extremely developed manufacturing
- High material quality demands to minimize parasitic absorption and non-radiative relaxations, high index $\rightarrow$ difficult light extraction

rare earth ions



- dipole-forbidden 4f-4f transitions"
- High internal quantum efficiency
- Long radiative lifetime (ms scale) → low baseline max cooling power

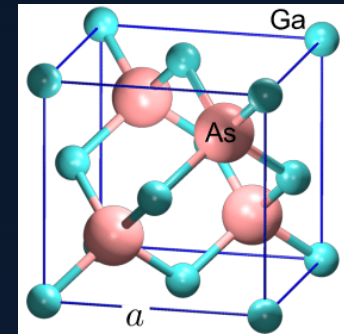
fluorescent dyes



rhodamine 101

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“molecular” systems

What are the limitations in ASF cooling?

$$\eta_c = \eta_{abs} \eta_{EQE} \frac{\bar{\nu}_f}{\nu_p} - 1$$

$$P_{cool,max} = \eta_p \eta_c h \nu_p \tilde{R} N_t$$

Must overcome several sources of “heating” and interface/optical/material inefficiencies

A) Materials growth/fabrication limited:

- 1) Parasitic absorption (background impurities) η_{abs}
- 2) Non-radiative transitions/recombination
- 3) High-density $N_t \sim 1$ quenching

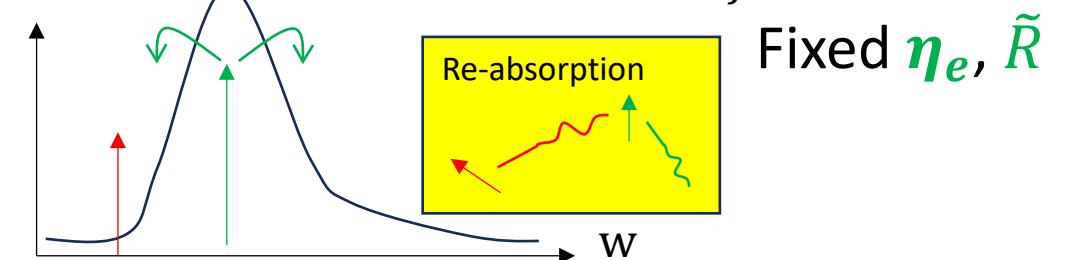
B) What can be “engineered” by photonic integration?

- 1) η_{EQE} depends on how many of the emitted heat-carrying **photons** **escape** η_e ($\eta_{EQE} = \frac{\eta_e W_{fl}}{\eta_e W_{fl} + W_{nr}}$)
- 2) $\bar{\nu}_f = \int (d\omega \dots)$ - **photon recycling** / spectral filtering allows “trapping” Stokes or “red-shifted” photons and efficiently out-coupling them at higher frequencies $\rightarrow$ **higher** η_c
- 3) **Radiative rate** $\tilde{R}$ can be enhanced via Purcell effect
 $\Rightarrow$ higher $\eta_{abs} \rightarrow 1$ at smaller $\nu_p \Rightarrow$ higher η_c
 $\Rightarrow$ higher $W_{fl} \Rightarrow$ higher $\eta_{EQE} \rightarrow 1 \Rightarrow$ higher η_c
- 4) Coupling efficiency η_p can be engineered to be nearly unity

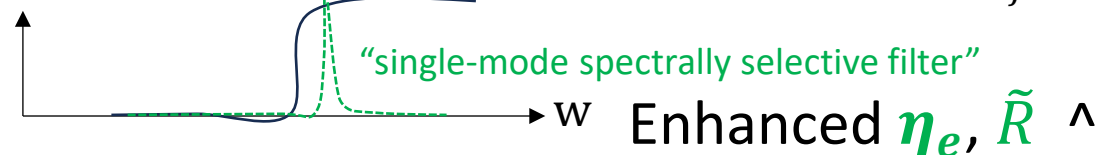
—Surface passivation, band bending (DHS engineering)

—**SMILES+FRET (chemistry)** increases N_t

Bulk: red-shifting reduces $\bar{\nu}_f$



Nanophotonic: “frustrate” red-shifting increase $\bar{\nu}_f$



1. **Background absorption:** Must be suppressed below $\alpha_r/100$ for net cooling
2. **EQE:** Must exceed ω_{in}/ω_f threshold at target temperature
3. **Si compatibility:** Determines manufacturing cost and integration scale
4. **Inverse design leverage:** Higher gain-per-iteration for photon-engineered structures

FY1: Yb:ZBLAN in optimized photonic cavity —push cooling threshold to 300 K
FY2: Organic QD on Si wafer —demonstrate chip-scale cooling at 5 W/cm²
FY3: GaAs EL+ LPC-PV stack —demonstrate net positive power budget



Platform	Peak EQE	Background Alpha_B	SI Integration	T Range	Inverse Design Leverage	TRL
Yb³⁺:ZBLAN	~99%	~10⁻⁴ cm⁻¹	Difficult	<119 K		5
Er³⁺:YAG	~95%	~10⁻³ cm⁻¹	Bonded	<200 K		4
Org. QDs / Dye	~85%	~10⁻² cm⁻¹	Spin-coat	260–310 K		3–4
GaAs LED EL	~99.7%	~0 (abrupt)	Wafer bond	~293 K		3

1. **Background absorption:** Must be suppressed below $\alpha_r/100$ for net cooling
2. **EQE:** Must exceed $\omega_{in}/\omega_{fthreshold}$ at target temperature
3. **Si compatibility:** Determines manufacturing cost and integration scale
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Laser Cooling Materials Explorer
Interactive analysis with real time pump wavelength tuning

Parameters

- Temperature (K): 300K
- Rank By: P_{cool_opt} (W/m²)
- # Scaling: 0.6e
- C Scaling: 4.0e
- A Penetration: 0.5e
- [Landscape View](#)

Power Density vs. η_{ASP}

Y-axis: P_{cool_opt} (W/m²) (log scale, 0.01 to 1000)
X-axis: η_{ASP} (%) (linear scale, 0 to 100)
Color scale: QF (0.1 to 1.0)
Size scale: A

Selected Material

Gallium arsenide antimonide (GaAsSb)

Bandgap (eV)	0.700
η_{ASP} (%)	28.2%
I_{ASP} (%)	75.0%
A (μm ²)	1.770e
A _p (μm ²)	2.201e
A _{pump} (cm ²)	4.4
A _p /V	1.00e-7
R _{cool} (%)	1.00e-7
C _{cool} (%)	6.00e-30
P _{cool_opt} (W/m ²)	0.6
I _{cool} (W/m ²)	1.00e-14
QF (%)	75.0%
I_{ASP}	2.50

[Load Interactive Spectra](#)

The screenshot displays the 'Laser Cooling Materials Explorer' interface. The top navigation bar includes the application name and a search bar. The left sidebar contains two main sections: 'Parameters' and 'Material Database'. The 'Parameters' section includes sliders for 'Temperature (K)' (set to 200), 'Pumping (W/cm²)' (set to 0.0), 'Fluorescence (nm)' (set to 635), 'Polarization' (set to 0.0), and a 'Generate Spectra' button. The 'Material Database' section lists various materials, with 'Sulfuric anhydride (SAR)' selected. The main area, titled 'Interactive Spectra', shows a plot of 'Fluorescence Intensity (a.u.)' versus 'Wavelength (nm)'. A yellow curve represents the 'Excitation' spectrum, and a blue curve represents the 'Emission' spectrum. A red box highlights the peak of the emission spectrum at approximately 635 nm. The right sidebar displays a table of properties for 'Sulfuric anhydride (SAR)', including 'Molar mass', 'Density', 'Melting point', 'Boiling point', 'Vapor pressure', 'Solubility', 'Refractive index', 'Thermal conductivity', 'Thermal expansion coefficient', and 'Thermal stability'.

Story – why integration?

$$\eta_c = \eta_{abs} \eta_{EQ} \frac{\bar{\nu}_f}{\nu_p} - 1$$

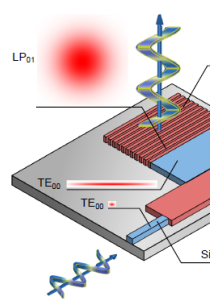
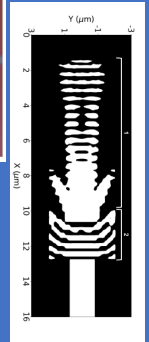
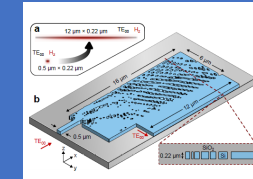
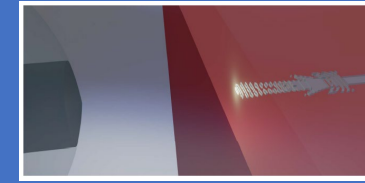
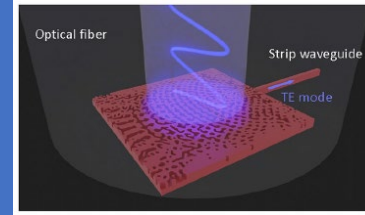
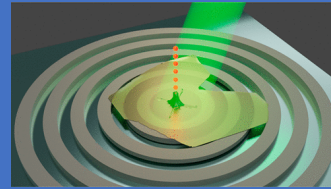
$$P_{cool,max} = \eta_p \eta_c h \nu_p \tilde{R} N_t$$

Show schematic of several bulk ASF experiments showing big cuvettes/bulky/large-etalon system



Show schematic of many integrated systems that Outperform their bulky counterparts (Raktim metasurfaces, etc)

Near-perfect coupling / absorbers



Purcell enhancement & Photon recycling

Dynamic Temporal & Spatial Targeting (sub-us and um scale)

Pictures/works of photonic networks that do multiplexing fast & local

$$\begin{aligned} \eta_p &> 95\% \\ \eta_e &> 90\% \\ \eta_c &\gtrsim 30\% (GaAs, R101: ethanol) \quad (\bar{\nu}_{f,recycl}) \\ P_{cool,max} &\gtrsim 1000s \text{ W/mm}^2 \quad (\tilde{R}_{Purcell} \sim 10\tilde{R}) \end{aligned}$$

Limitations?

$$\eta_p \ll 1\%$$

$$\eta_e \sim 10\%$$

$$P_{cool,max} < W/mm^2$$

$$\eta_c \sim 1\% (Yttb: SiO_2) - 10\% (Cr: LiSaF)$$

Strong red-shifting (large absorption path)

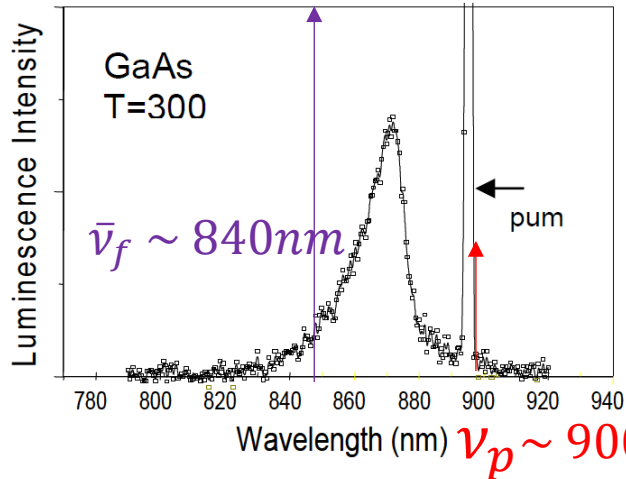
A “few-mode” spectrally selective thin-film cold plate solves many issues:

- Higher mean fluorescence
- Reduced optical losses (low coupling losses & high pump absorption)
- Reduced red-shifting (heating)
- Radiative rate enhancement via high-Q/V /field confinement Purcell enhancement means higher absorption efficiency and EQE
- Reduced entropy (higher thermodynamic efficiency) – allows going beyond Carnot limit

GaAs

$$\eta_q = \frac{\tilde{\nu}_f - \nu_p}{\nu_p}$$

$$p_{cool,opt} \sim \frac{\eta_c}{1 + \eta_c} \left(\eta_q - \frac{\alpha_b}{\alpha} \right)^2 (\eta_e \tilde{R})^3 h \tilde{\nu}_f$$



$$\eta_c = \frac{\bar{\nu}_f}{\nu_p} \sim 10\%$$

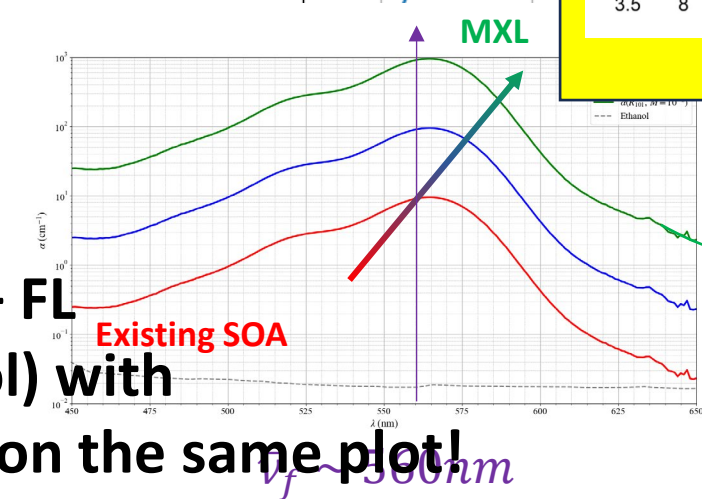
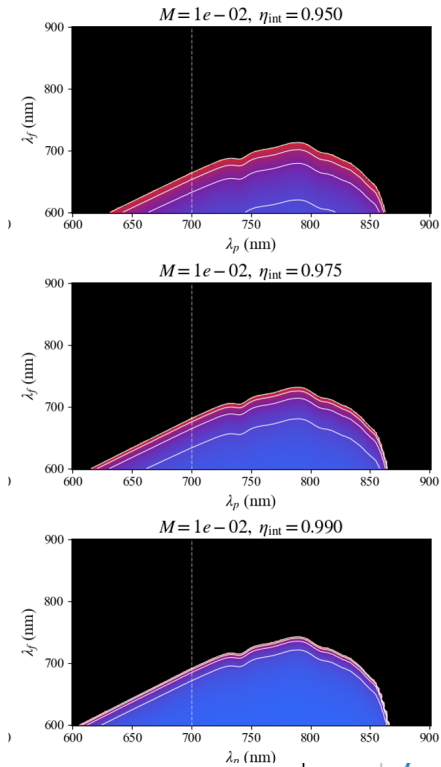
(Photon recycling)

$$P_{cool,max} \sim (\eta_e \tilde{R})^3 \sim 10000 \text{ W/mm}^2$$

(Purcell ~ 10x)

TODO: absorption + FL spectra (incl ethanol) with diff concentrations on the same plot!

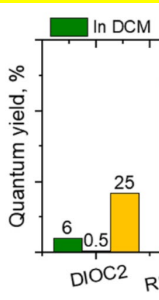
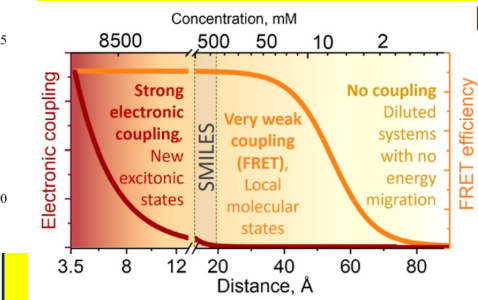
R101:ethanol



$$P_{cool,max} = \eta_c h \nu_p \tilde{R} N_t \sim 10000 \text{ W/mm}^2$$

(Purcell > 10x)

Higher density (“concentration”) ~ 0.1M. How? FRET SMILES



$$\eta_c \frac{\bar{\nu}_f}{\nu_p} \sim 30 - 40\%$$

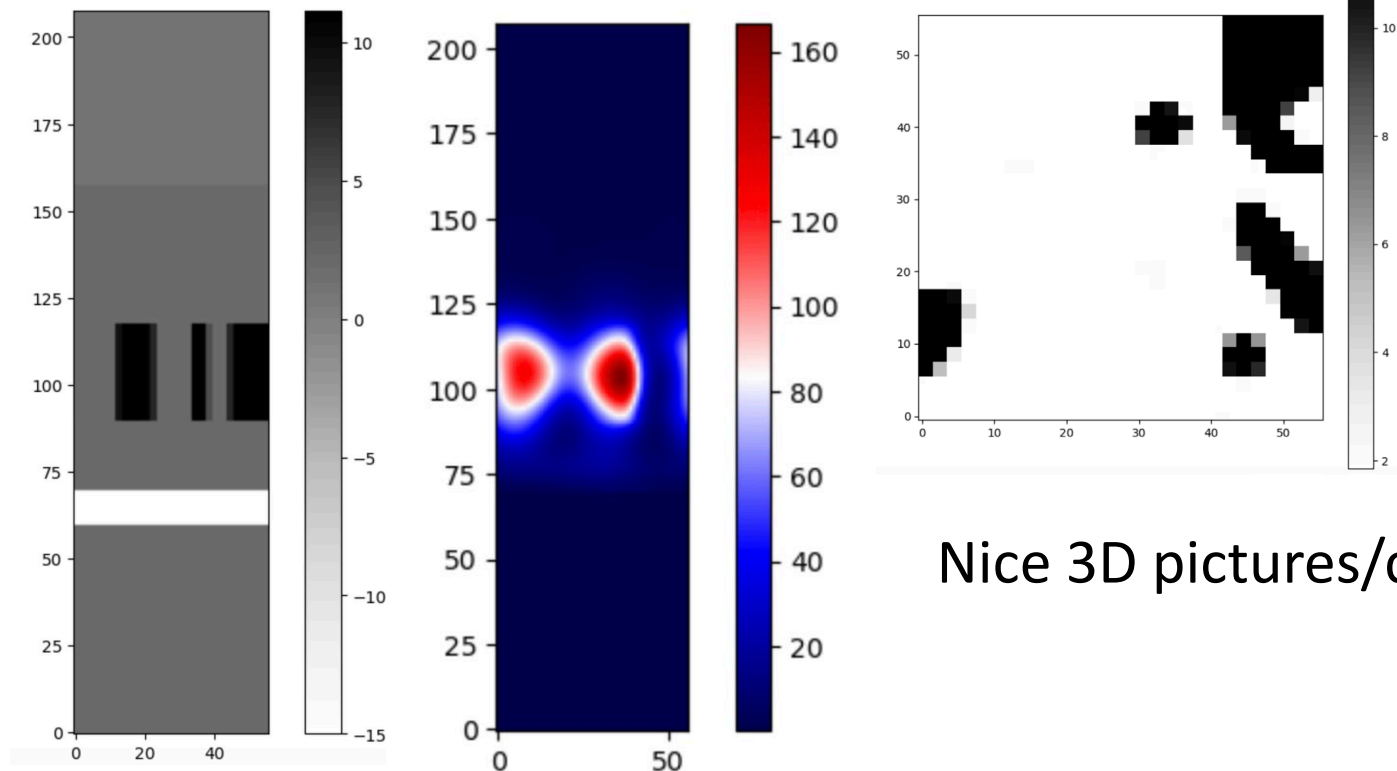
(Photon recycling)

Preliminary Metasurface Model

First: background on inverse design

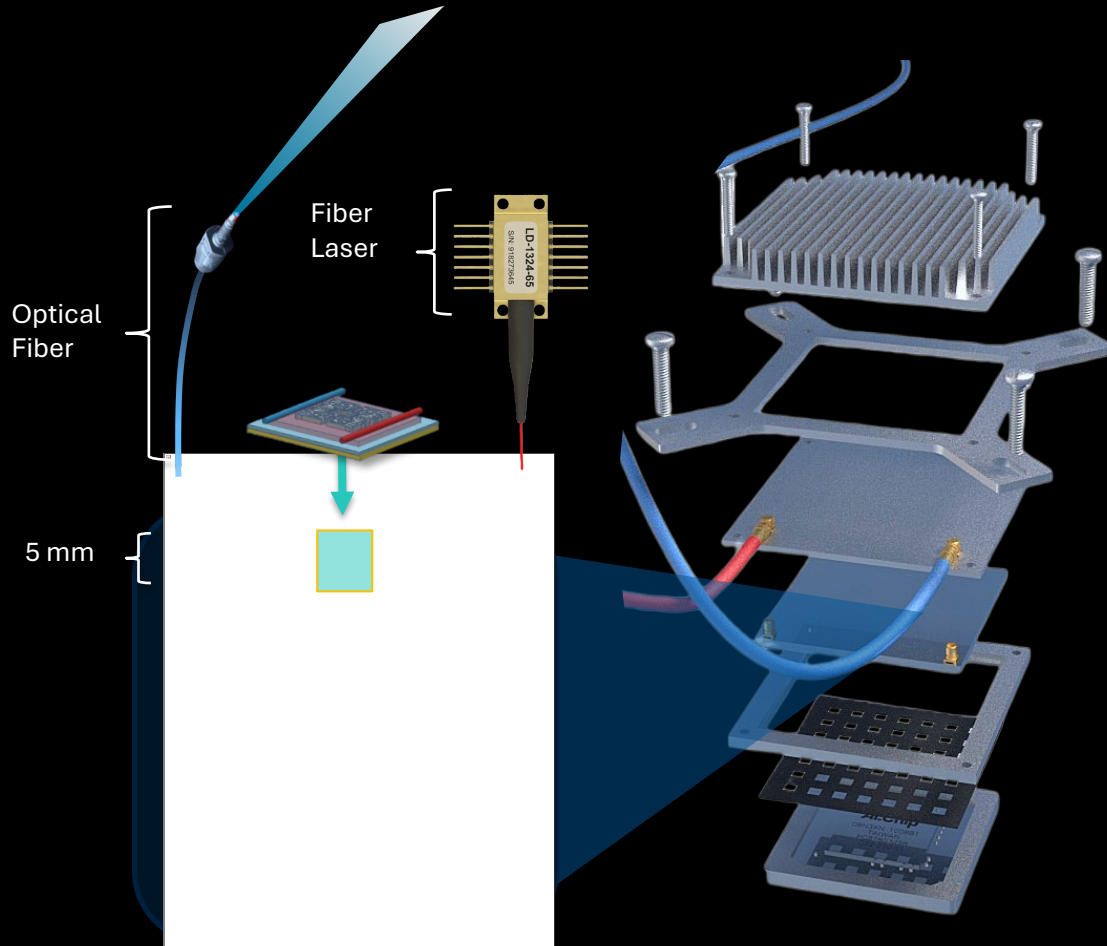
Second: (3D model and field profile, spectrum of your

$$f(\rho) = \frac{1}{2} \omega_p \epsilon_0 \operatorname{Im} [\chi_{p,l}] \left(\int_{design} (1 - \rho) |\mathbf{E}_p|^2 dv + \int_{liquid\ on\ top} |\mathbf{E}_p|^2 dv \right)$$



Nice 3D pictures/cad!

Commercialization Roadmap >> MVP-02 | Early Access



Key Takeaways

- COP of 1.69: Every 60W used lifts 100W optically
- Fiber integrated power recovery @ 70% efficiency
- Cr:LiSAF extractor @ 33% efficiency
- Integrated Photonic network
- Dynamic spatial / temporal targeting

$$P_{chip}: 1000W$$

$$P_{total}: 4100W$$

$$\eta_{aSF}: 0.33$$

$$\eta_{LPC}: 0.87$$

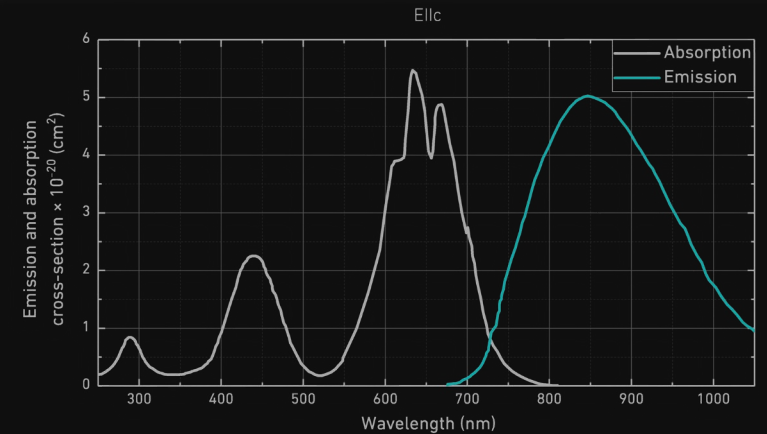
$$\eta_{laser}: 0.74$$

$$P_{used} = [1 - \eta_{LPC}(1 + \eta_{aSF})\eta_{laser}]P_{total} = 590W$$

$$P_{removed} = \eta_{aSF}\eta_{laser}P_{total} = 1000W$$

$$\eta_{PCP} = \frac{P_{removed}}{P_{used}}$$

$$\frac{1000W}{590W} = 1.69$$



References:

<https://doi.org/10.1002/adom.202403222>

[https://doi.org/10.1016/0038-1098\(92\)90289-L](https://doi.org/10.1016/0038-1098(92)90289-L)

Conclusions & Next Steps

✓ KEY CONCLUSIONS

- 1 **Closed-Form Optimum Exists:** The optimal MR strength s^* has a clean analytical expression —no iterative solver needed for control. (Include Resistance conclusions)
- 2 **Hot Spots Change Everything:** The 2D α/β model shows MR must engage at much lower chip-average fluxes than 1D predicts. 3D stacked ICs have no viable alternative.
- 3 **Photonic Beats Thermoelectric:** Anti-Stokes luminescence is contact-free, scalable to sub-100 μm , and can achieve COP>1 via photon entropy pumping.
- 4 **LPC Makes It Energy-Positive:** The removed heat becomes electricity via the LPC-PV stack —reducing data center PUE rather than increasing it.
- 5 **Inverse Design is the Enabler:** Adjoint topology optimization unlocks near-unity EQE, directional emission, and spectral matching —the three factors that cross the cooling threshold.

🏠 NEXT STEPS

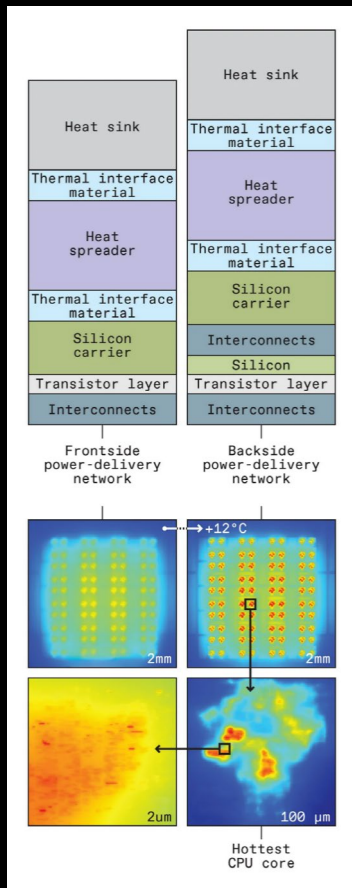
- S1 Build 2D spatial MR simulator with adjoint-coupled photonic design for Yb:ZBLAN
- S2 Fab organic QD thin film on Si test chip —measure EQE and net cooling at 300 K
- S3 Integrate GaAs LED + spectral filter + GaInP PV —measure net P_{elec} output from 5 W/cm² heat source
- S4 Extend 2D model to 3D for stacked die architecture —partner with TSMC CoWoS team

❓ OPEN QUESTIONS

- Can background absorption be engineered below 10⁻⁵ cm⁻¹ at 300 K in a solid-state host?
- What is the practical reliability lifetime of anti-Stokes luminescent films at 5 W/cm² pump intensity?
- Does the LPC-PV stack recover sufficient power to justify the stack complexity at the 5nm node?

THE GOAL

"Make the Thermal Wall Transparent —
and Extract Power from the Waste Heat"



FEATURE COMPUTING

YOU CAN COOL CHIPS WITH LASERS?!?!

Startup plans to cool data centers by converting heat to light

BY JACOB BALMA ALEJANDRO M. RODRIGUEZ
16 OCT 2025 | 10 MIN READ

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